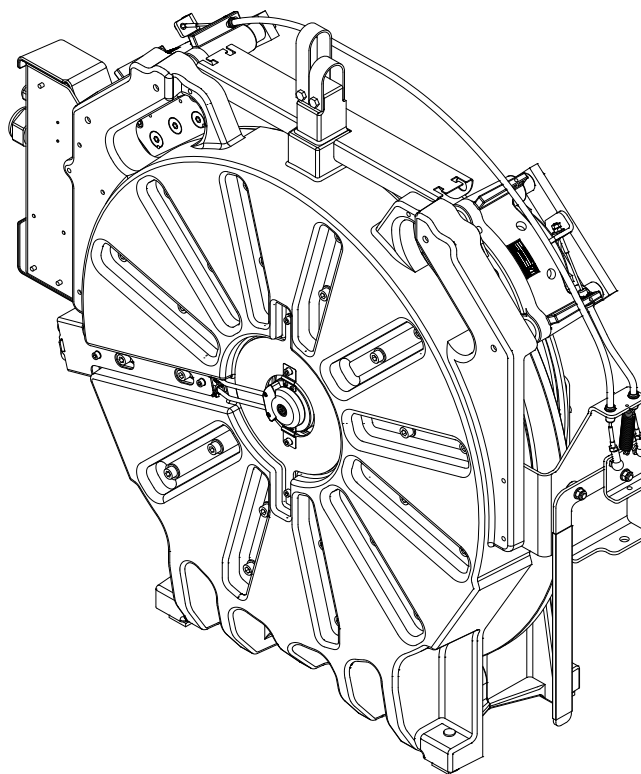


MX14 MACHINE, ROTOR AND BEARING, REPLACEMENT



C8040D0C

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Original instructions

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1 ABOUT THIS INFORMATION

This information describes the bearing and rotor replacement methods and actions for the MX14 machine.

Related information

- [Online version of this publication in Technical Information Portal](#)

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1.1 Scope

In scope

- Maintenance method
- Tools and materials for the work
- Method-related safety issues

Out of scope

- Local variations
- Safety principles and guidelines covered in generic safety instructions
- General guidelines for environmental management

Related information

- [Generic safety instructions for elevator maintenance \(7\)](#)
- [For more information on environmental issues, refer to QD-01559 Environmental Excellence - Field Operations in KONE Way Document Repository](#)

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1.2 Audience

This information is intended to be used by people who are competent and familiar with elevator maintenance and have received proper training on methods and safety as specified by KONE.

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1.3 Feedback

To report errors in the information or suggest improvements, select one of the following:

- Send email to ktd@kone.com.
- Respond to the questionnaire here: <https://forms.office.com/r/efjC5Pktf8>

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1.4 Translations

If you need assistance in getting technical information translated, contact ktd@kone.com.

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1.5 Approvals and changes

Issue: A

Date: 2025-03-13

Reference CR: CR-181406

Changes:

- Added new topic [Safety measures \(70\)](#).
- Added new topic [Remove rotor \(71\)](#).
- Added new topic [Install new rotor \(85\)](#).

Compiled by:

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Approved by:

- Tony Enholm, Unit Maintenance Director

Translation approved by:

dc01246c

1.6 Signs and symbols

 922f9911	Risk of electric shock
	Electrostatic discharge hazard
 c1d2d6fb	Danger of magnetic field
	Suspended load fall hazard
	Crushing hazard
	General mandatory action
	Wear eye protection

	Wear protective clothing
	Wear task-related safety gloves
	Wear head protection
	Wear safety shoes
 b6facb24	Install safety fence
 05c81d54	Two persons required to complete task

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1.7 Generic safety instructions for elevator maintenance

The following safety instructions support the product-specific and component-specific elevator maintenance information:

- AM-01.03.001 Preventing Falls from Height when Working on Elevators, Escalators and Building Doors
- AM-01.03.002 Take 5 - Electrical Safety when Working on Elevators, Escalators and Autowalks
- AM-01.03.009 General Instructions for Material Handling, Using Transportation and Lifting Equipment
- AS-01.03.101 Elevator Maintenance, Safety Manual
- AS-01.03.103 Hosting Contractors and Visitors on Elevator Maintenance and Installation Sites
- AS-01.05.006 Safe Access to Deep Elevator Pits
- AS-12.02.001 Electrostatic Discharge (ESD) Protection for Field Personnel
- QD-01559 Environmental Excellence - Field Operations
- TC04406E01-EN Electrostatic Discharge (ESD) Protection on the Field, Core Learning Guide

Related information

- [*AM-01.03.001 Preventing Falls from Height when Working on Elevators, Escalators and Building Doors*](#)
- [*AM-01.03.002 Take 5 - Electrical Safety when Working on Elevators, Escalators and Autowalks*](#)
- [*AM-01.03.009 General Instructions for Material Handling, Using Transportation and Lifting Equipment*](#)
- [*AS-01.03.101 Elevator Maintenance, Safety Manual*](#)
- [*AS-01.03.103 Hosting Contractors and Visitors on Elevator Maintenance and Installation Sites*](#)
- [*AS-01.05.006 Safe Access to Deep Elevator Pits*](#)
- [*AS-12.02.001 Electrostatic Discharge \(ESD\) Protection for Field Personnel*](#)
- [*For more information on environmental issues, refer to QD-01559 Environmental Excellence - Field Operations in KONE Way Document Repository*](#)

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2 MX14 MACHINE

The MX14 machine is gearless, synchronous, permanent magnet axial machine. It is equipped with two independently working brakes.

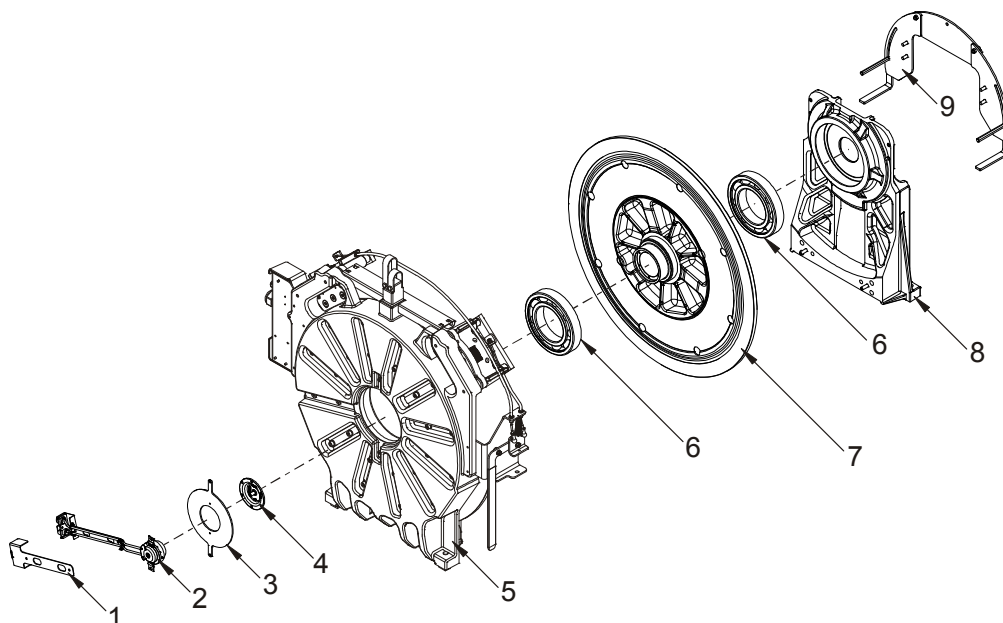
MX14 machine is designed to be used in both machine-room-less (KONE MonoSpace®) and machine room (KONE MiniSpace™) solutions for mid-range residential and office buildings.

The MX14 machine uses the following bearings:

- KM50052525 Bearing 6226-2RS1 / C3VT189F2 (Position 6 in the illustration below)

Parts of the MX14 machine described in this document include:

1. Encoder wire cover plate
2. Encoder
3. Cover plate
4. Encoder flange
5. Stator assembly with body1
6. Lower bearing (left), top bearing (right)
7. Traction sheave
8. Body2
9. Rope guards



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Related information

- [KM50052525 Bearing Ball D230/130MM W40MM 6226-2RS1 / C3VT189F2](#)

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3 SAFETY AND MAINTENANCE TARGETS

3.1 Maintenance method safety

Refer to *AS-01.03.101 Elevator and Building Door Maintenance - Safety Manual* for general safety precautions and personal safety issues.

Note the following safety items when performing the maintenance work described in this instruction:

- General KONE safety regulations must be followed during the work. All normal safety measures required in elevator maintenance must be implemented. Local safety regulations must also be followed.
- Refer to *AM-01.03.002 Take 5 - Electrical safety when working on elevators, escalators and autowalks*. The Take 5 safety initiative is designed for installation, servicing, maintenance and modernization work done on the elevators.
- **Make sure you have adequate information to do the work and fully understand all the possible effects of your actions.**
 - **IF IN DOUBT - ASK.**
- Talk to your supervisor if you have any doubts about the working method or safety related issues.
- Any local variations to the method must be supported by a local method statement and risk assessment.
- Site survey must be done before starting maintenance work.
- If there is any risk to building occupants from unguarded landing entrances, consider using safety fences and other safety measures to protect the area around of topmost landing door.
- Use gloves when handling ropes, in order to avoid injuries from rope splinters.
- Always use the safety harness attached to an approved fixing point when working in the elevator shaft.
- Condition and load rating of lifting equipment must be checked before moving the MX machine.
- Loading capacities must be checked before moving the MX20 or NMX20 machine (for example lifting points, and floors).
- Take special care when lifting heavy components like the machines. When loosening any fixings, make sure that the remaining supports can carry all the loads bearing on them.
- The blocking pin of the car blocking device must be in the bottom slot of the blocking plate before checking or adjusting the mechanical brake.
- If MRL elevator there is a risk of high pressure oil leakage when using the hydraulic pump. Wear eye protection.
- Be careful when opening fixing screws and nuts to avoid dropping metal items into the pulley housings. Although the housings are protected with covers and brushes, it is possible that small particles get caught in the pulley grooves.
- Door of the maintenance access panel must always kept closed. Even, when the main power is OFF, or when not working with the panel.
- Prevent the reconnection of the possible emergency power supplies etc.

WARNING:

MX motor will act as a generator, when the motor is rotated for example manually.

Make sure that the motor wires are insulated and cover is fitted to motor terminal box and that you cannot come into contact with exposed conductors.



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WARNING:

Inverter drives usually remain energized for about 5 minutes after the power has been disconnected.

DO NOT work on the drive, hoisting motor or braking resistors until you have verified that this energy has been discharged.

Test equipment must be set to the 1000 VDC range. The test equipment must be checked before and after the test to make sure that it functions correctly.



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Related information

- [AS-01.03.101 Elevator and Building Door Maintenance - Safety Manual](#)
- [AM-01.03.002 Take 5 - Electrical Safety when Working on Elevators, Escalators and Autowalks](#)

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3.2 Personal protective equipment (PPE) for elevator maintenance

All maintenance work must follow local KONE unit safety rules.

Use the following personal protective equipment (PPE) in all elevator maintenance:

- Cut resistant gloves against sharp objects
- Working clothes
- Safety shoes with toe guard

Use the following PPE always when your local KONE unit requires:

- Head protection, choose one of the following (refer to local safety rules):
 - Bump cap
 - Safety helmet

Where needed, use the following PPE to protect against specific hazards:

- Safety goggles (refer to local safety rules).
- Protective gloves against chemical hazards, cold and heat (refer to local safety rules).
- Hearing protection

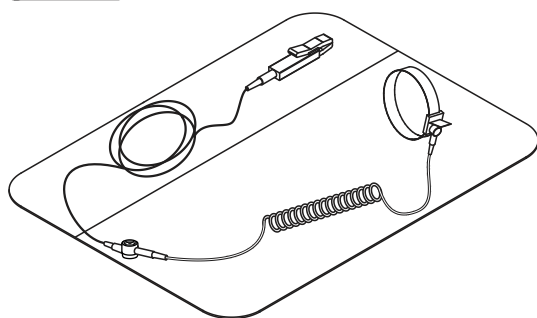
- Dust mask
- Eye protection against UV radiation
- Fall restraint or arrest equipment.



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3.3 Basic rules for preventing ESD hazards

Follow the procedures and use the tools described in AS-12.02.001 *Electrostatic discharge (ESD) protection for field personnel*.



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1. Earth yourself before touching components, use a wrist strap combined with a portable ESD working pad.
2. Disconnect all power supplies.
3. Do not unpack the spare parts before installation.
4. Do not touch the components on the board unnecessarily.
5. Hold the PC-board from its edges do not touch the connectors.
6. The old component must be handled in the same way as the new one.
7. Use ESD - working pad to lay the components down (Insulation material does not protect against ESD damage).

8. Do not wear combinations of clothing that can create static charges.

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3.4 Safety components

3.4.1 Traceability of safety components

KONE as an economic operator (installer, manufacturer, distributor and importer) keeps a record of suppliers and customers for each installed and replaced safety component. If needed, this enables finding and organizing an immediate replacement of a faulty safety component in any elevator.

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3.4.2 Identification of safety components

KONE safety components can be identified as follows:

- KONE safety components are marked with a CE marking label.

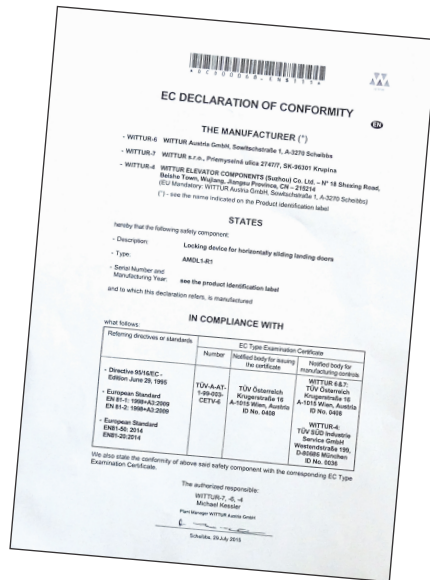


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- Safety components have a declaration of conformity document (available in electronic format and sometimes delivered with the safety component).



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3.4.3 Replace safety components

A list of the elevator's safety components is available as a part of the elevator's technical documentation in KONE PDM.

Safety Component List



D8 - KONE Elevator

EN-81-20:2014

Certificate No NL12-400-1002-002-61
 KEN 44438285
 SAP Sales Orders 350643328
 Customer Name
 Unit Street AHVENISTONTIE 24
 Unit City HÄMEENLINNA
 Country FI - Finland
 Unit Building Name ASSI, AHVENISTON SAIRAALA HÄMEENLINNA
 Rated load.....(kg) 2000
 Rated speed.....[m/s] 1.6
 Elevator: number of floors 5



Installed / delivered safety components

Brake (Acop and UCMP)

KM	Description	S/N	Certificates	KONE Documents
KM51535468V020	BRAKE	C00SR16235069=20124		
KM51535468V020	BRAKE	C00SR16235072=20124		

Buffer

KM	Description	S/N	Certificates	KONE Documents
KM51172694	BUFFER	23OB0751-01-02-03	EQSCH 001/4	CT-05.01.012/en/Q.1 CT-05.01.383/en/G.1

Car door

KM	Description	S/N	Certificates	KONE Documents
KM601282C01	DOOR	WAT231701678 (F3407E32023317016784)	TÜV-A-AT-1-99-0007-FKTV-7	CT-03.01.212/en/G.1

Landing door

7085ff76

1. Verify that the new spare part has a CE marking.
2. Replace the component.
3. Perform the commissioning tests according to EN 81-1 or EN 81-20 and component specific AS-documents.
4. If the declaration of conformity document is delivered with the spare part, store it in the elevator's logbook folder on-site.
5. Mark the safety component replacement in the elevator's logbook.
6. Backreport the replacement with KONE Field Mobility (KFM) or in a locally agreed way.
7. Demolish the old part beyond repair before disposal.

6524896b

3.5 Magnetic field

The machines are surrounded by a strong magnetic field when the motor body is opened and the rotor uncovered. Because of the magnetic field, electronic devices may malfunction or cease to operate in the vicinity of the MX machines.

WARNING:



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STRONG magnetic attraction force!

If you use a pacemaker, a hearing aid or other kind of sensitive electronic device keep away from the working area.

Be careful when working or handling metallic objects in vicinity of open rotor

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3.6 Working with MX machines

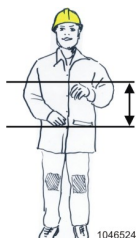
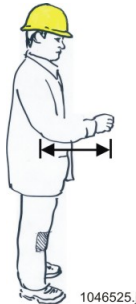
Because of the construction of the MX-machines, note the following:

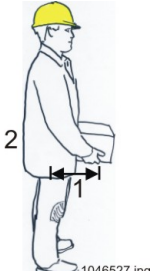
- Machine is a permanent magnet AC-motor.
- Because of this feature magnetizable dust from sources such as welding, grinding, gravel or similar must be kept outside the machine.
- When there is a danger of magnetizable dust getting inside the machine, use always a reinforced adhesive tape to cover the air gap between the rotor and the machine frame.
- Do not drill any holes to the motor body (drilling is allowed only to the bars on the body sides).

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3.7 Recommended working postures and ergonomic methods

These recommended working postures and ergonomic methods have been identified to contribute a better workplace and your well-being.

Working height Adjust height of working to enable your hands to be within the vertical area of chest and belt buckle.	 1046524.jpg 7b7d51ed
Working distance Adjust working distance to enable the object to be within length of forearms.	 1046525.jpg e731ca52

Lifting Focus on the lift, even if the lifted object is light! 1. Go close to the lifted object. 2. Use your legs. 3. Keep your back untwisted and straight. 4. Keep both feet on the same lane.	 7f4c66e8
Carrying 1. Keep the lifted object as close to the body as possible. 2. Keep your back untwisted and straight.	 f1402df9
Loading of transport cart Slide the component onto the KONE transport cart instead of lifting it. Moving heavy objects without a handle If the object doesn't contain a handle, push/slide the object instead of pulling.	 af4d8d31  aed99f48

NOTE:

If work is being carried out when kneeling, with bent back/legs or with hands elevated above chest or object is further than length of forearms, ensure to limit time of exposure and alternate the working postures. Make sure that the working area is un-obstructed and not hindering you.

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3.8 Transportation



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The transportation of heavy parts should be planned in advance (case by case).

Some repair tool boxes weigh more than 60 kg. Use site trolleys to transport the tool boxes.

When it is difficult to transport tool boxes into the topmost landing or there is no space for tool boxes at the topmost landing, store them as near as possible. Use safety fences, hoardings and warning signs as required.

NOTE:

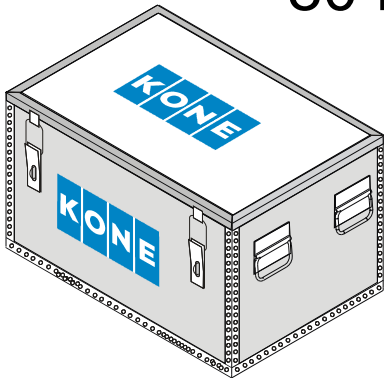
Keep the emergency exits clear.

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4 TOOLS

4.1 Tool box for MX14 installation (KM51105493V000)

80 kg



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Bearing replacement tools

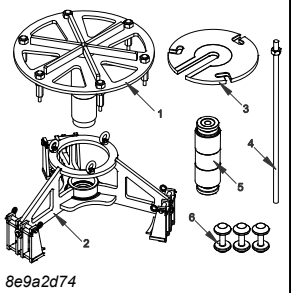
- Dimension: 680 mm × 750 mm × 620 mm

CAUTION:

Transportation of the tool box is done by two qualified persons considering its weight.

Table 1: Contents of tool box for MX14 installation

No.	Name / Description	Order number / Picture
1	Lower extractor tool (for supporting the machine)	KM51083506V000
2	Top extractor tool	KM51082942V000
3	Bearing removal plate	
4	M20 threaded bar	
5	Bearing removal bushing	KM51089652V000
6	Bearing removal claws	



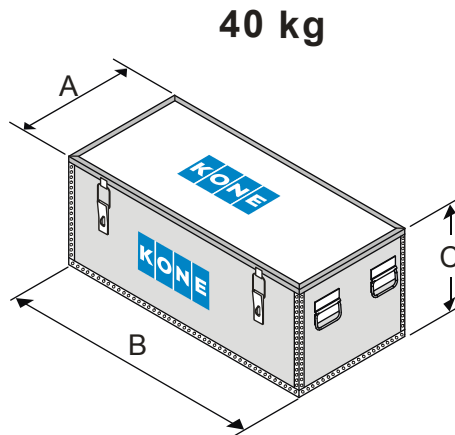
Related information

- [AS-01.05.007 Maintenance tool catalog](#)
- [AS-04.06.001 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room-less Elevator](#)
- [AS-04.06.002 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room Elevator](#)

33c8d708

4.2 MX18 seal replacement toolset 1 (KM1376117)

EXAMPLE:



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A = 550 mm

B = 580 mm

C = 416 mm

KM number	Description	Pcs
KM853851G01	Tool box made of plywood	1
KM237404	Metric combination spanner 30 mm	1
KM1375803	Metric combination spanner 27 mm	1
KM237388	Metric combination spanner 24 mm	1
KM237362	Metric combination spanner 19 mm	2
KM237347	Metric combination spanner 17 mm	2
KM237321	Metric combination spanner 13 mm	2
KM237297	Metric combination spanner 10 mm	2
KM1375881	Metric socket wrench 30 mm ($\frac{3}{4}$ "	1
KM1375947	Metric socket wrench 24 mm ($\frac{3}{4}$ "	1
KM1375806	Metric socket wrench 30 mm ($\frac{1}{2}$ "	2
KM1375936	Metric socket wrench 24 mm ($\frac{1}{2}$ "	2
KM1375938	Metric socket wrench 19 mm ($\frac{1}{2}$ "	2

KM number	Description	Pcs
KM1375939	Metric socket wrench 17 mm (½")	2
KM1375844	Metric torque wrench (½") 70 - 340 Nm	1
KM1375845	Metric torque wrench (½") 40 - 200 Nm	1
KM1375842	Pry bar, large (820 mm)	1
KM1375843	Pry bar, small (430 mm)	1
KM846587H01	Chisel, bent tip pry bar	1
KM1375811	Metric socket wrench 13 mm (½")	2
KM1375812	Hex key socket 14 mm (½")	2
KM1375813	Hex key socket 10 mm (½")	2
KM1375940	Hex key socket 8 mm (½")	2
KM1375941	Hex key socket 6 mm (½")	2
KM50014160	Hex key socket 17 mm (¾")	2
KM50014161	Hex key bit 17 mm	2
KM1375942	Ratchet (½")	2
KM1375943	Short ratched extension (½")	1
KM1375818	Long ratchet extension (½")	1
KM865272	Ratchet (¾")	1
KM167015	Multi grip pliers	1
KM1375944	Large diagonal cutting pliers	1
KM1375945	Small diagonal cutting pliers	1
KM1375822	Hammer	1
KM1375821	Rubber hammer, Nupaflex	1
KM1375888	Heat gun (230 VAC / 50 Hz) Excluded from USK sets.	2

KM number	Description	Pcs
KM1375868	KM wrench	1
KM1376679	Screwdriver (insulated) 1.6 × 10.0	1
KM1376678	Screwdriver (insulated) 1.2 × 8.0	1
KM1376674	Screwdriver (insulated) 1.2 × 6.5	1
KM1375825	Screwdriver (insulated) 1.0 × 5.5	1
KM1375826	Screwdriver (insulated) 0.8 × 4.0	1
KM1375827	Screwdriver PH2	1
KM269113	Metric hex key set 1.5 - 10.0 mm	2
KM1375828	Wire stripper pliers	1
KM100941	Knife	1
KM1375829	Scissors	1
KM132118	Meter wheel, 3 m	1
KM230532	Flat file (half rough)	1
KM269115	Wiper	1
KM1375830	Circlip pliers 40 - 100 mm	1
KM1375831	Screw tap driver M5-M12 mm	1
KM1375832	Screw tap set M6	1
KM1375833	Screw tap set M8	1
KM1375834	Screw tap set M10	1
KM1375835	Screw tap set M12	1
KM1375836	Screw tap set M20	1
KM142109	Feeler gage set 0.05 - 1.00 mm	1
KM1375837	Feeler gage belt 0.1 mm	1

KM number	Description	Pcs
KM1375838	Chargeable lamp (230 VAC / 50 Hz) Excluded from USK sets.	1
KM1375869	Brush 25 mm	4
KM1375839	Compressor (for grease tube)	1
KM1375873	Compressor (for silicon tube)	1
KM1375840	Refuse sack 150 l / roll	1
KM1375841	Cable extension 10 m (EU socket) Excluded from USK sets.	3
KM225672	Mandrel 6 mm	1

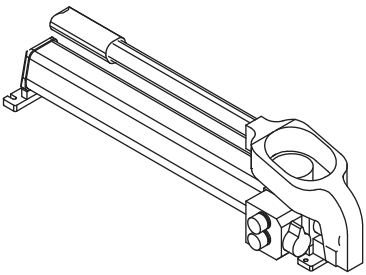
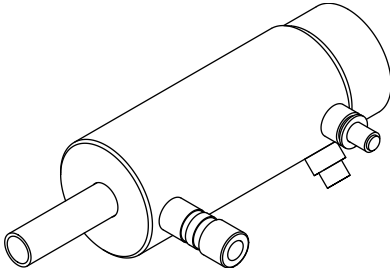
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4.3 MX18 seal replacement tool set 2 (KM1376118) - partially needed

Chain hoist, 3pcs Vital VR3-15 (1,5 t) KM1366021  06f6d2df	Load chain, 1pcs KM856396G02 L = 480 mm Testing certificates in tool box 3
Load chain, 1pcs KM856396G01 L = 280 mm Hoisting device testing in accordance to local regulations.	

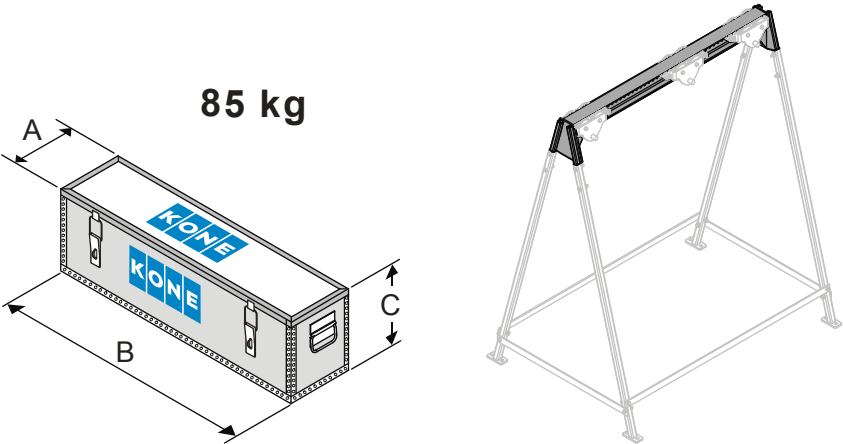
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4.4 MX18 seal replacement tool set 3 (KM1376119) - partially needed

Hydraulic pump, 1pcs KM846589G01	Hydraulic double acting ram, 1pcs KM846588G01
 14d89fc0	NOTE: Ram must have piston (KM840302H01) included.
	 0a0d3b6c

7b4a22ae

4.5 MX18 seal replacement toolset 5 (KM1376530)

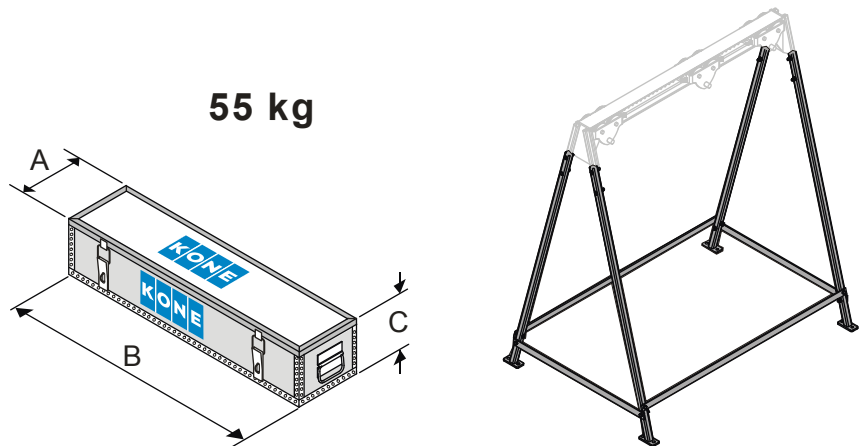


A = 380 mm
B = 1780 mm
C = 486 mm

Tool box made of plywood, 1pcs KM856766G01	Assembly drawing 846594
Beam, 1pcs KM846595G01	

1861b5d4

4.6 Tool set 6 (KM1376531)

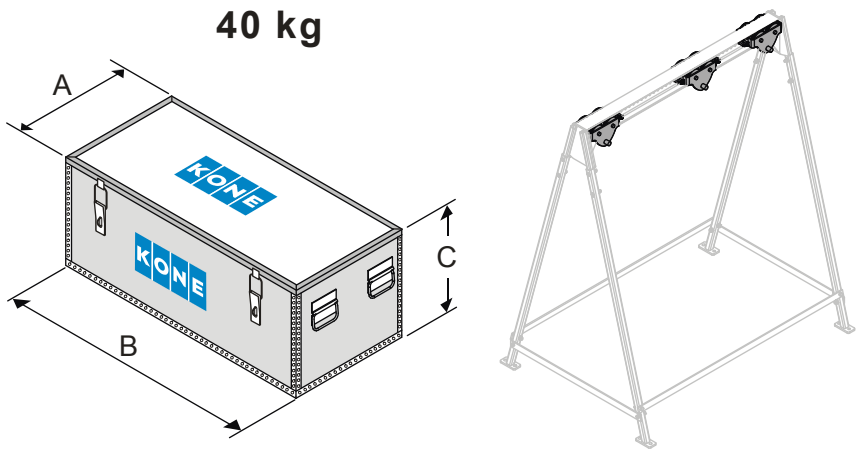


A = 330 mm
B = 1950 mm
C = 166 mm

Tool box made of plywood, 1pcs KM856767G01	A-frame leg, 2pcs KM846597G02
A-frame leg, 2pcs KM846597G01	Reinforcing bar, 2pcs KM846597H03
Reinforcing bar, 2pcs KM846597H04	Bolt, 8pcs M12x25 DIN7991-8.8A3F KM1366013
Bolt, 16pcs M12x60 DIN7991-8.8A3F KM1366014	Spring washer, 24pcs M12 ST A3F DIN127 ITE1017335
Washer, 16pcs A13 DIN125 A3F KM277230	Nut, 24pcs M12 DIN934-8A3F KM277220

279a4f44

4.7 Tool set 7 (KM1376532)

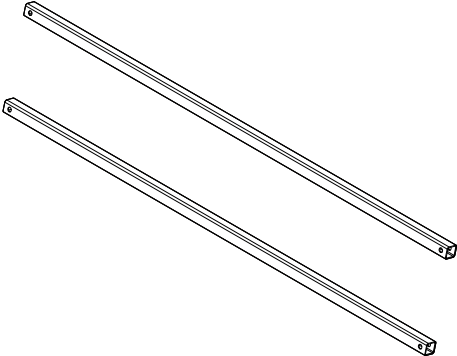
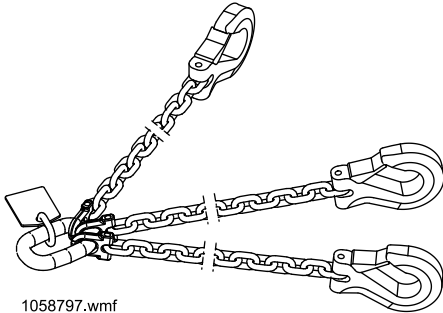


A = 380 mm
B = 580 mm
C = 265 mm

Carriage, 3pcs KM1366015	Tool box made of plywood, 1pcs KM856768G01
Axle tube, 3pcs D30/d27x95 KM1366016	Closure pin, 3pcs KM856392G01
Ring pin 4X27, 3pcs DIN11023 Attached to closure pin KM923859	

5c26a1d5

4.8 Locally sourced tools

Side beams for A-stand KM961602  2ecaa371	Load chain, 3-branch Erlatek VIP-VHM / VML3-1M or equivalent L= 1.0 m, adjustable, capacity = 630 kg  1058797.wmf d09e83a8
Rubber mats and dampers For floor protection. – Minimum size required: 1000 mm × 1000 mm	Plywood pieces For covering, separated stator assembly. – Minimum size required: 1000 mm × 1000 mm × 2 mm Also used as an alternative solution for floor protection.
Wood beams For supporting, rotor on side beams. – Minimum size required: 1200 mm × 40 mm × 40 mm	Cleaning spray Würth Pro-Clean or equivalent
Silicone glue (black) Würth 0892-317-6 or equivalent For encoder cover plate, assembly.	Hex socket head screw M10 × 25 For stator screw replacement, if needed.

b5eac207

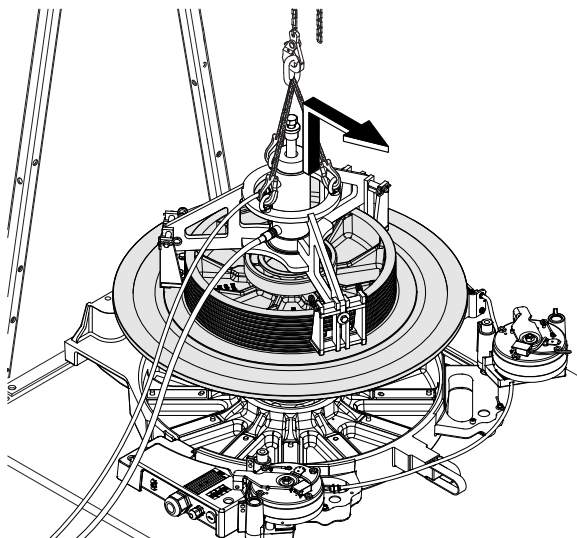
5 BEARING REPLACEMENT



05c81d54

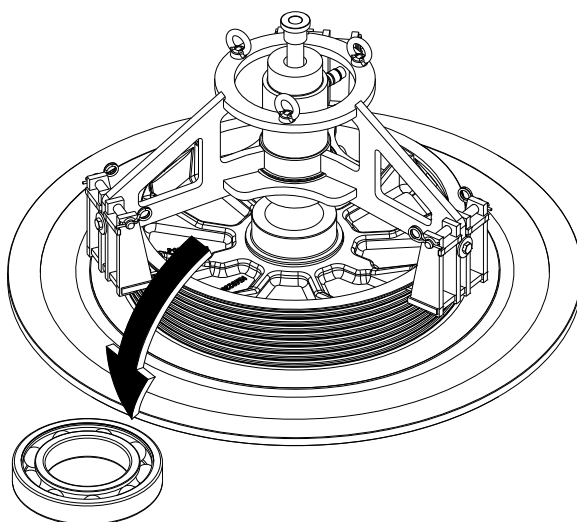
The main phases of bearing replacement for MX14 machine are:

1. Disassembling the rotor.



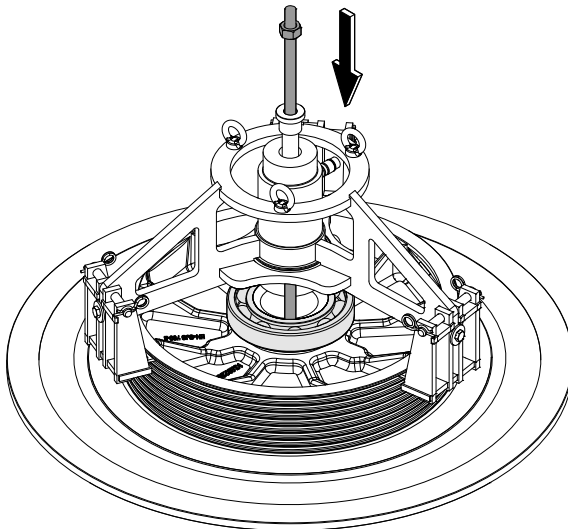
83b3d173

2. Removing the bearings.



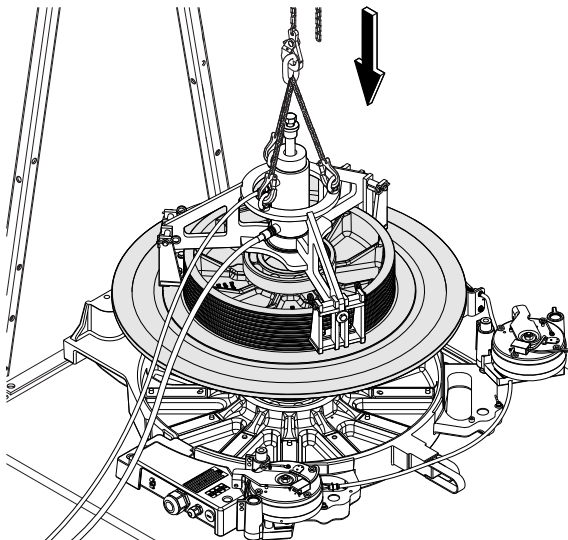
dc33ce43

3. Installing the bearings.



e2909181

4. Assembling the rotor.



00eb14ed

5055662e

5.1 Safety measures

The following safety measures must be carried out before the actual work.



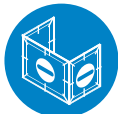
1. Protect the floor surface if needed.

Use rubber mats, dampers or plywood pieces.

2. Install the A-stand.

Fix the chain hoist to the A-stand.

3. Secure the working area with safety fences.



b6facb24

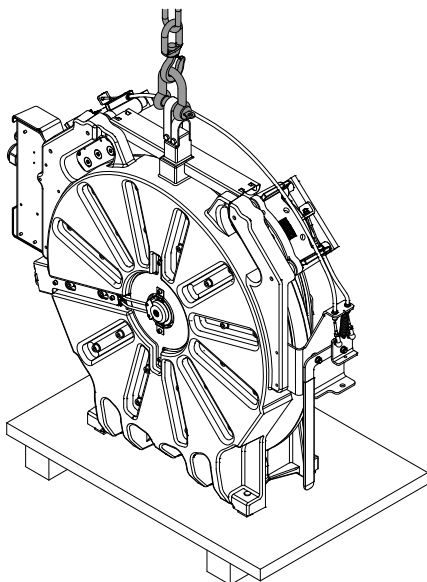
4. Check that all required equipment are on the work site.
5. Make sure that the A-stand modification is done correctly.

For more information, refer to related information.

NOTE:

Do not attach side beams at this point.

-
6. Always keep the chain hoist attached when the machinery is in vertical position and when working with rotor assembly detached from the machinery.



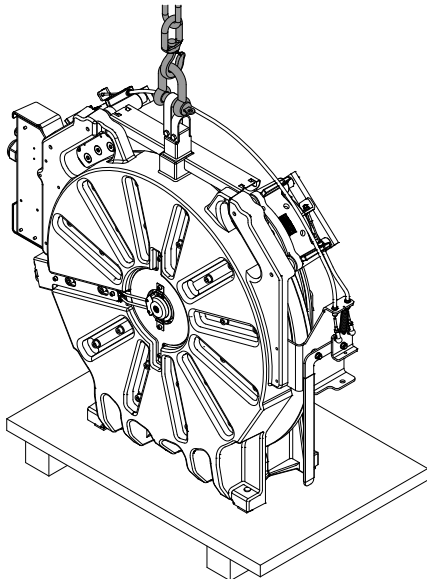
607ad7e9

d5f4e406

5.2 Disassemble rotor

5.2.1 Disassemble encoder

1. Remove MX14 machine from machine room or elevator shaft.



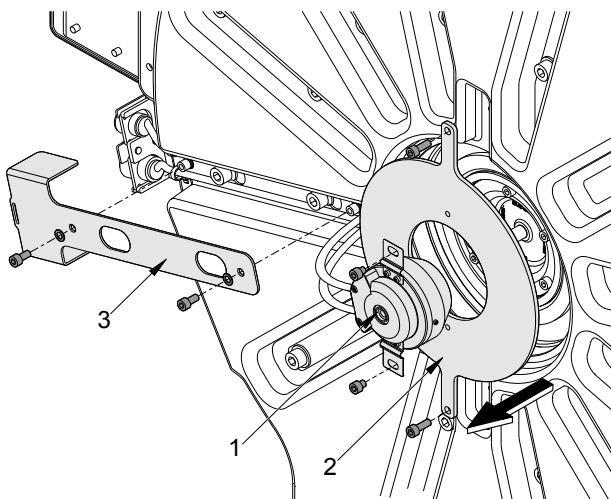
607ad7e9

- For more information on elevator with machine room, refer to the related information.
- For more information on machine-room-less elevator, refer to the related information.

NOTE:

Always keep the chain hoist attached when the machinery is in vertical position.

2. Detach the encoder (1). Remove the cover plate (2) and the encoder wire cover plate (3).

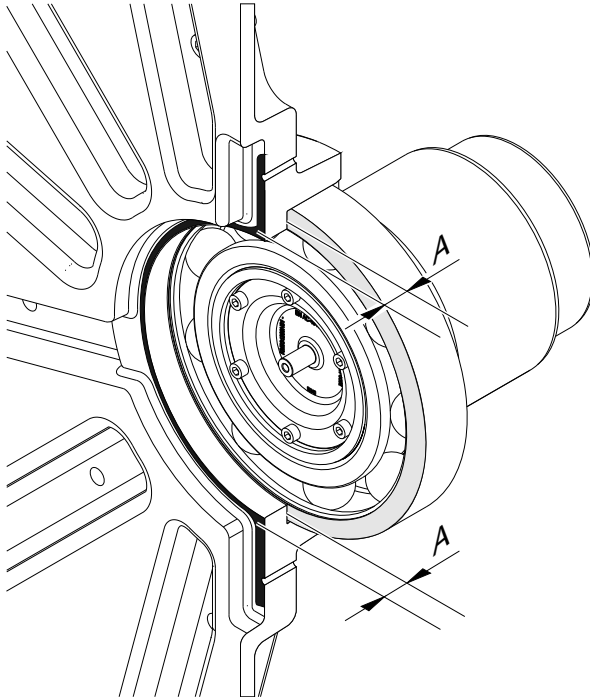


ff0ab1fb

NOTE:

Check that all the threads on the removed screws are intact. If not, check the threads on the machine. If machine threads are damaged, the whole machine must be replaced.

3. Remove all black silicone residues from the cast surface. Measure distance A.



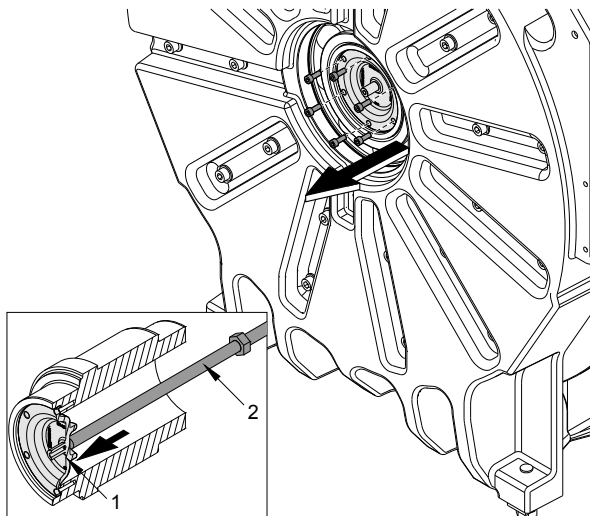
d5ee209f

- Check that the bearing surface is in line with the center axle top surface.
 - Write down the upper and lower results for later replacement of the bearing.
-

NOTE:

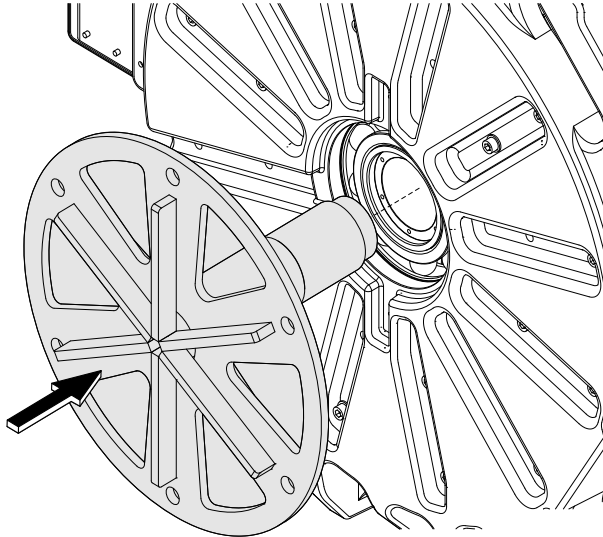
Pay attention not to measure distance “A” from the bearing shield.

4. Remove the encode flange (1).



6d016989

- Use the threaded bar (2) for example.
 - Insert the threaded bar (2) from the body.
 - Tap the end with a hammer to loosen the flange (1).
5. Check that the extractor tool bushings fit the axle elevator shaft.

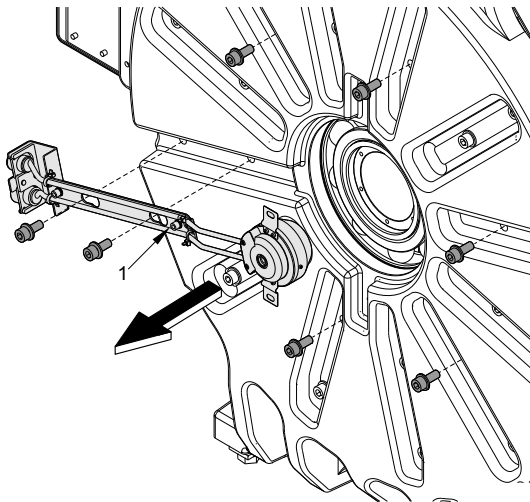


57342582

NOTE:

The whole machine must be replaced if the bushings do not fit the axle elevator shaft.

6. Remove the encoder.



0c60e248

- Remove the wire holder plate screws (1) and hex socket screws. (Counting from the encode wire plate screws, every other hex socket screw is to be removed.)

NOTE:

Check that all the threads on the removed screws are intact. If not, check the threads on the machine. If machine threads are damaged, the whole machine must be replaced.

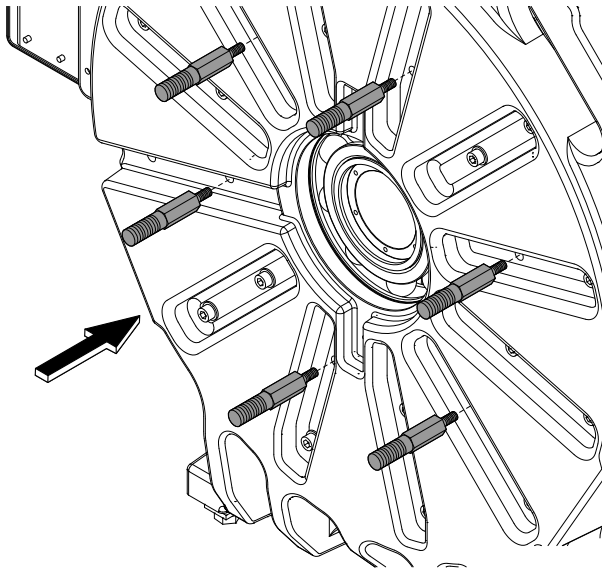
Related information

- [AS-04.06.001 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room-less Elevator](#)
- [AS-04.06.002 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room Elevator](#)

55874c3f

5.2.2 Install lower extractor tool

1. Install the six threaded bars to the open threaded holes.

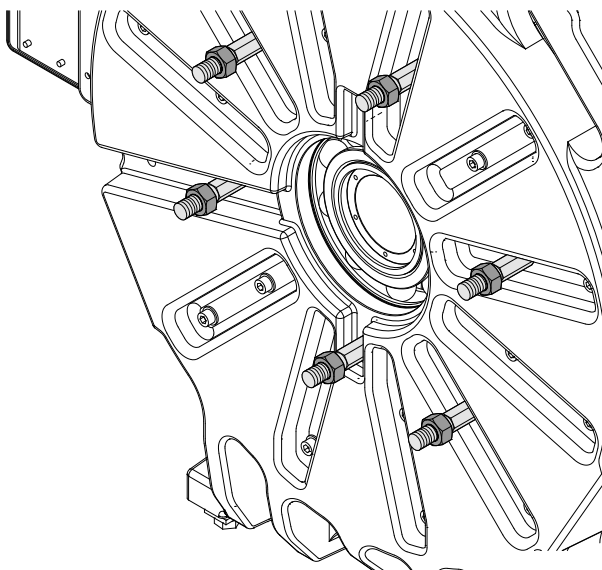


bd5b11f0

NOTE:

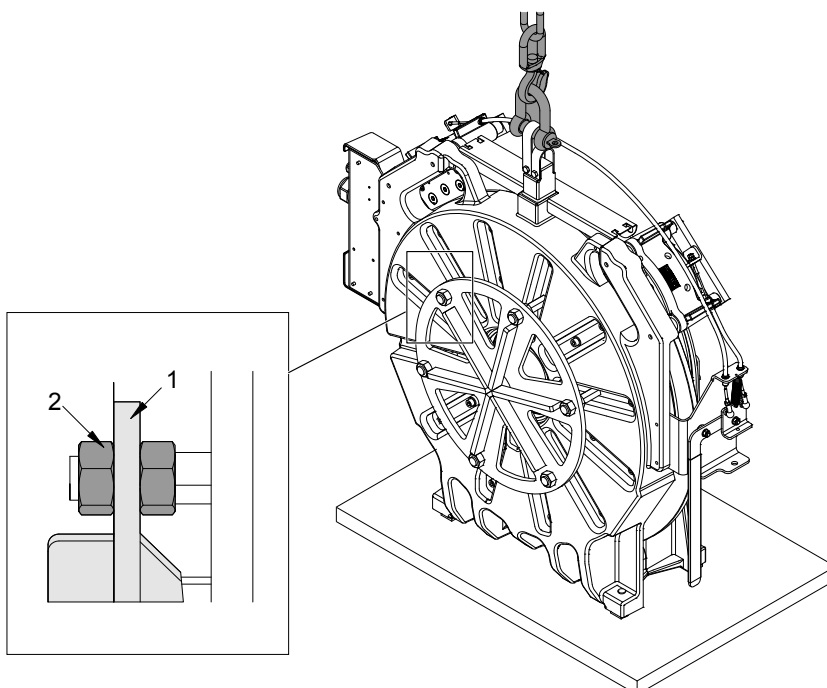
If removed screws are damaged, do not insert threaded bars.

2. Screw a nut to each of the six threaded bars.



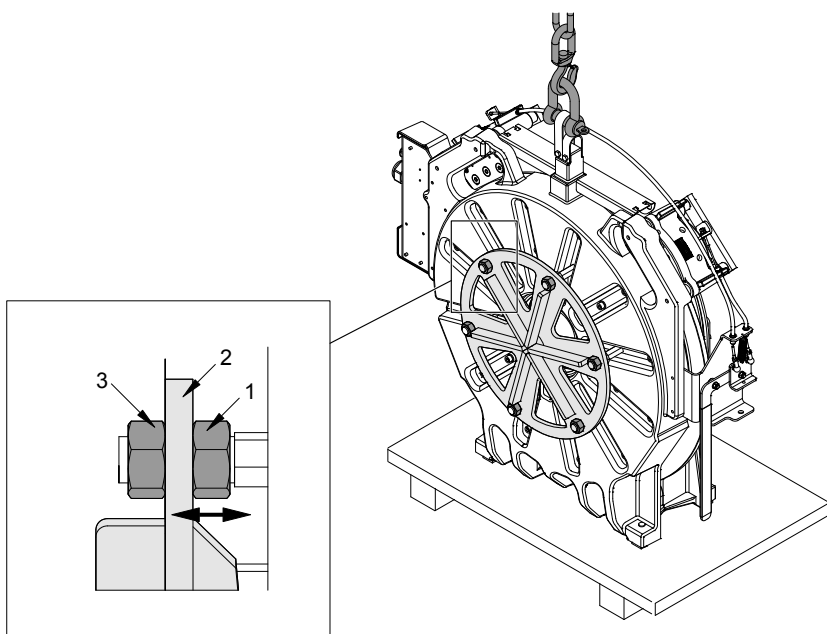
36607be4

3. Attach the lower extractor tool (1) to the six threaded bars and screw an outer nut (2) to each of the threaded bars.



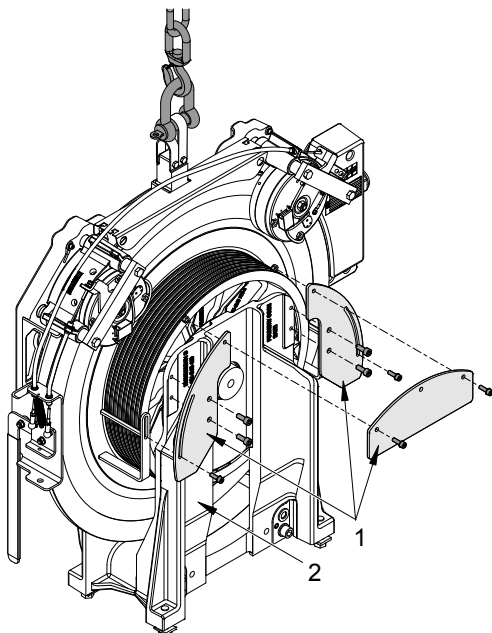
48440fd5

4. Align the lower extractor tool (2) with the nuts (1). Tighten the outer nuts (3) with a wrench.



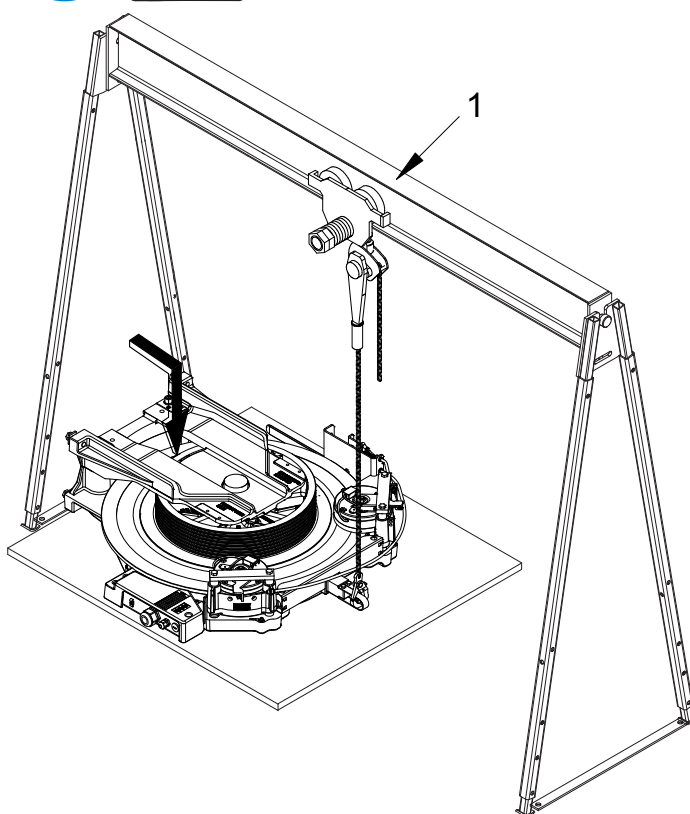
1c387198

5. Remove the rope guards (1) from body2 (2).



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6. Use A-stand (1) and chain hoist (2) to lift and tilt the machine until it is laid on a plywood piece or rubber mat / pallet.



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- With body 2 facing up and body 1 facing down
- With the installed lower extractor tool between the plywood piece or rubber mat / pallet and body 1

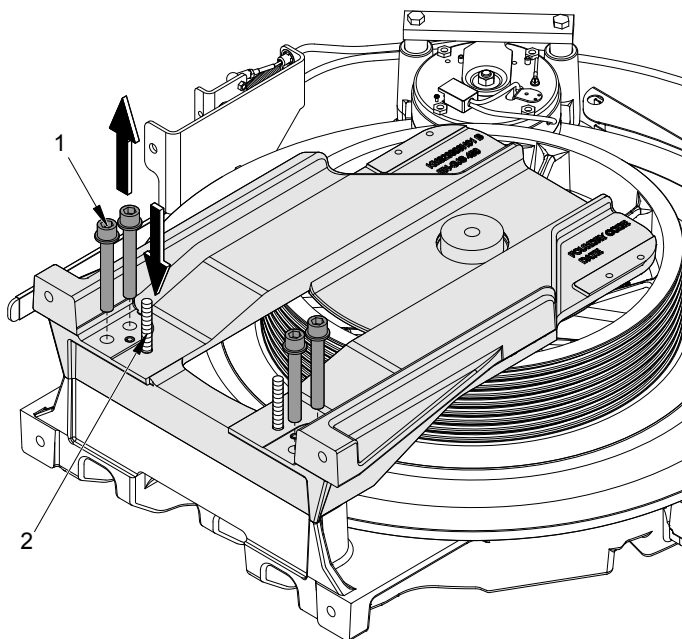
NOTE:

Place the machine on one end of the A-stand (1), not in the middle of it.

6126c3b4

5.2.3 Install top extractor tool

1. Remove the four screws (1) on body2 (two screws on each side). Insert two bolts (2) into the two threaded holes.



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NOTE:

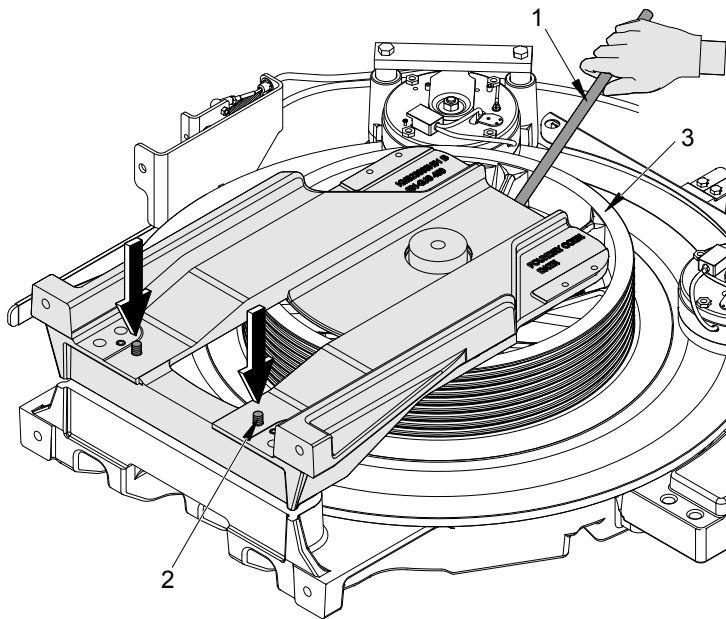
Before inserting bolts (2), check that there is a chamfer at the start of each threaded hole.

NOTE:

Use well-greased bolts.

2. Use an aluminum pry bar (1) to remove the body2.





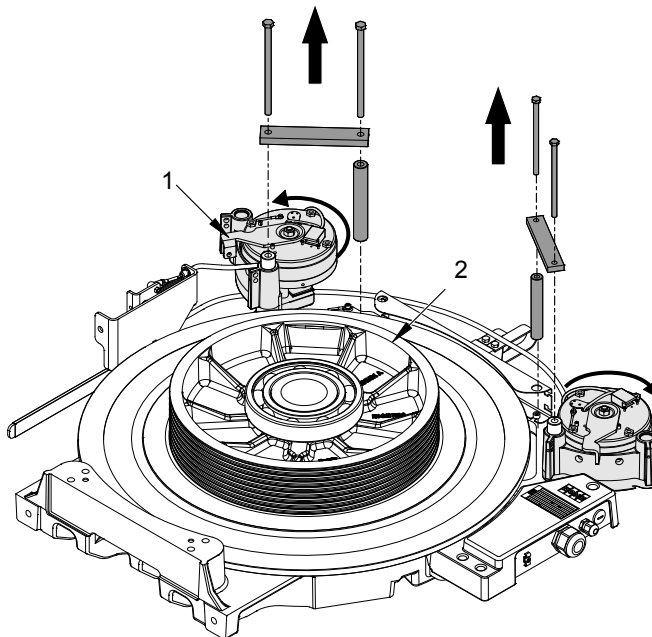
9f144981

- Use pry bar (1) and push out the other end of body 2 by tightening bolts (2) in turn so that body 2 is pushed upwards.

CAUTION:

Use the aluminum pry bar (1) with care to avoid damaging the traction sheave (3).

3. Move brakes (1) aside from traction sheave (2).
For more information, refer to related information.

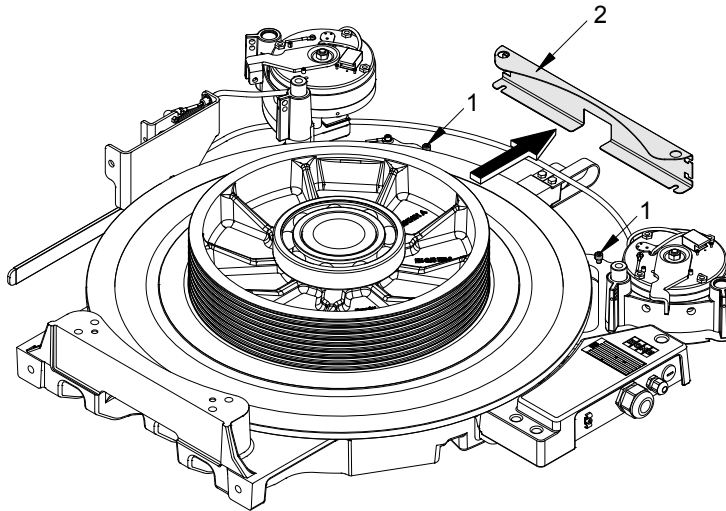


60b93f9c

NOTE:

Do not lift the brakes (1) away. Only turn it aside of traction sheave (2).

4. Loosen the cover plate screws (1) and slide off the cover plate (2).

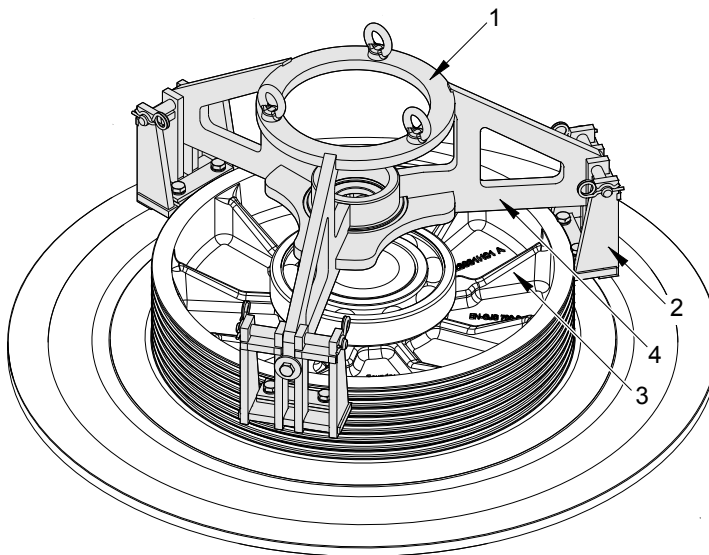


28309267

NOTE:

Do not remove the screws (1) completely.

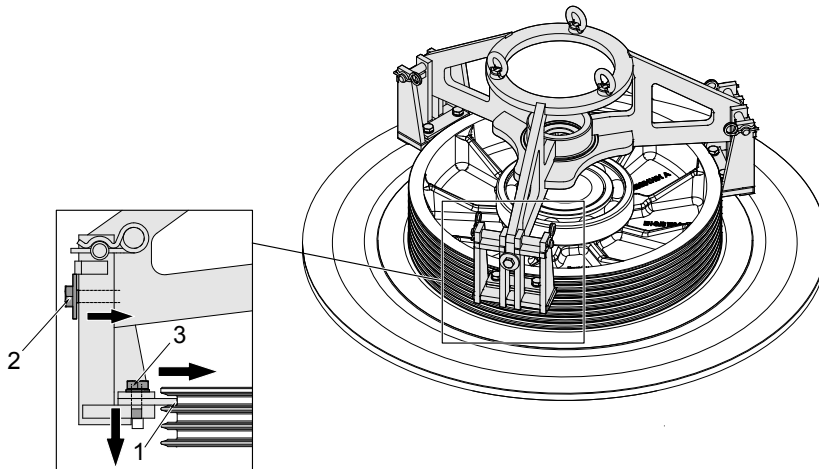
5. Install the top extractor tool (1) to the traction sheave grooves.



d393b80a

- Use the upmost groove to attach claws (2).
- Adjust the top extractor tool (1) so that the traction sheave ribs (3) are aligned with the wings (4) of the tool.

6. Check that the claws (1) reach the bottom of the groove undercut. Tighten the claw holder screws (2).



- Adjust claw depth if needed by screws (3).

NOTE:

Tighten the claw holder screws (2) evenly so that the tool is centered with the traction sheave.

Related information

- [Remove brake unit \(96\)](#)

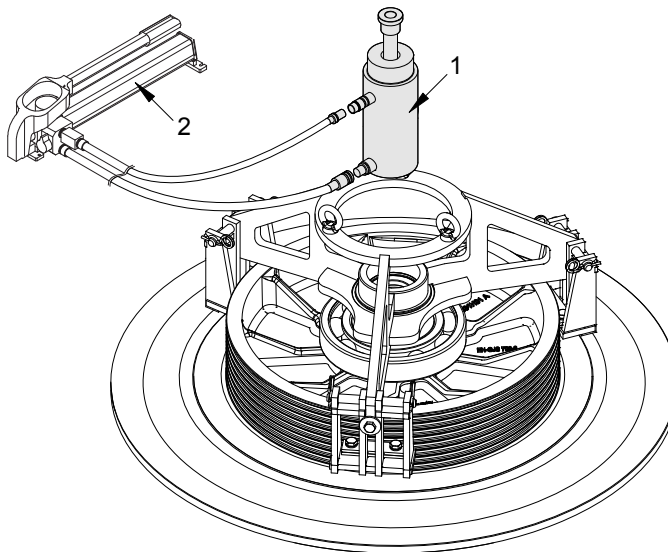
986e0c0d

5.2.4 Disassemble rotor assembly

1. Install the hydraulic jack (1) and connect the hand pump (2).

For more information on using the hydraulic pump, refer to related information.





5305bb91

WARNING:

Always wear eye protection when using a hydraulic jack.

WARNING:

The rotor has a strong magnetic field and a strong attraction force. Handle the rotor carefully and take care not to get your hands crushed between the magnet and the tools.

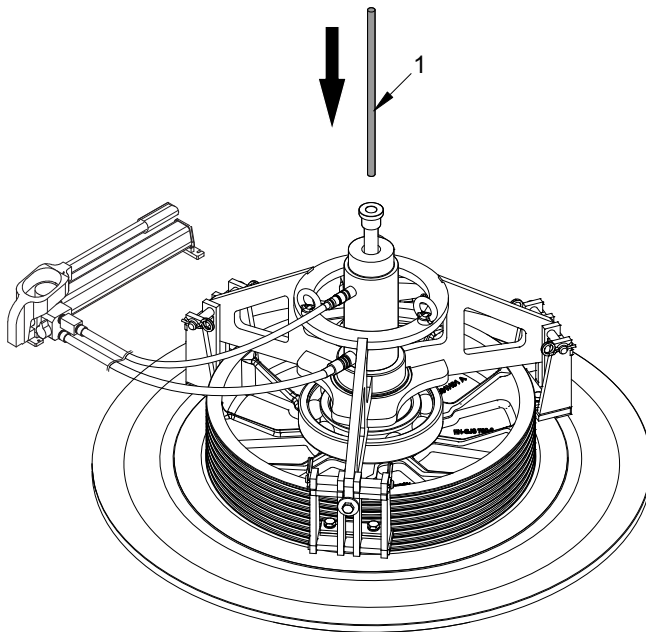
NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

2. Check tool alignment by inserting the M20 threaded bar (1).



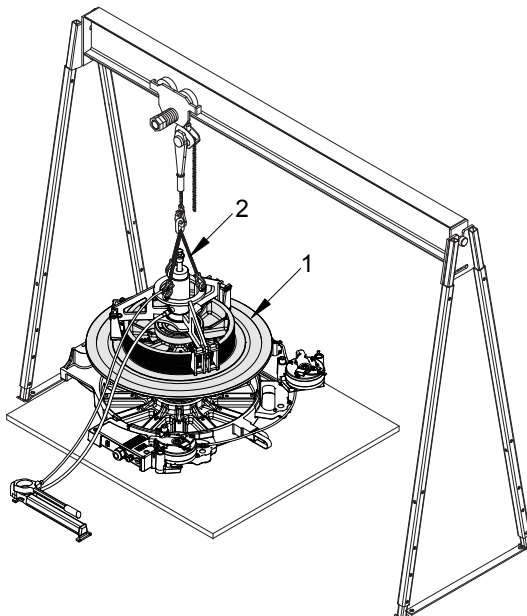
c1d2d6fb



e8939c97

Adjust if necessary.

3. Pump pressure until rotor assembly (1) becomes loose. Lift the rotor assembly (1) while pumping it up.



da3cd135

NOTE:

Always wear eye protection when using a hydraulic jack.

CAUTION:

Before pumping, always check that the piston has enough room to move for the operation to be done. Constantly look at the piston to make sure it does not collide unintentionally while pumping.

NOTE:

Once the pressure reaches 300 bar on the manometer, stop pumping immediately.

NOTE:

Adjust the 3-branch load chain (2) so that the lifted rotor assembly (1) does not exceed the height where the side beams can be attached.

NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

4. Lift and move rotor assembly (1) to the other end of the A-stand. Place plywood cover (2) on top of stator assembly (3) to protect it against dropping objects.



ec1cdd9d



WARNING:

Working under a suspended load is not allowed. Only replace bearings when chains are attached and beams are installed under the rotor assembly.

NOTE:

Rotor assembly with lifting tools weighs approximately 200 kg. Consider weight when lifting.

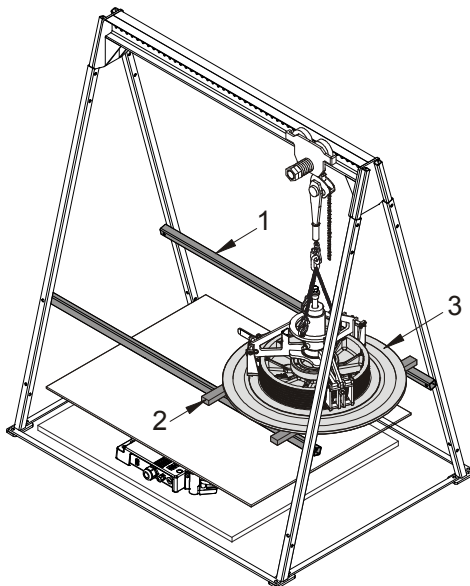
NOTE:

Do not remove shim plates (4). If shim plates (4) stick to rotor assembly, return shim plates (4) to stator assembly.

5. Install side beams (1). Place wood beams (2) on side beams (1). Lower rotor assembly (3) onto the wood beams (2).



ec1cdd9d



999be23e

NOTE:

Working under a suspended load is not allowed. Only replace bearings when chains are attached and beams are installed under the rotor assembly.

CAUTION:

Do not place rotor assembly (3) directly on side beams (1) to avoid damage of magnets.

NOTE:

Do not remove the 3-branch load chain at this point.

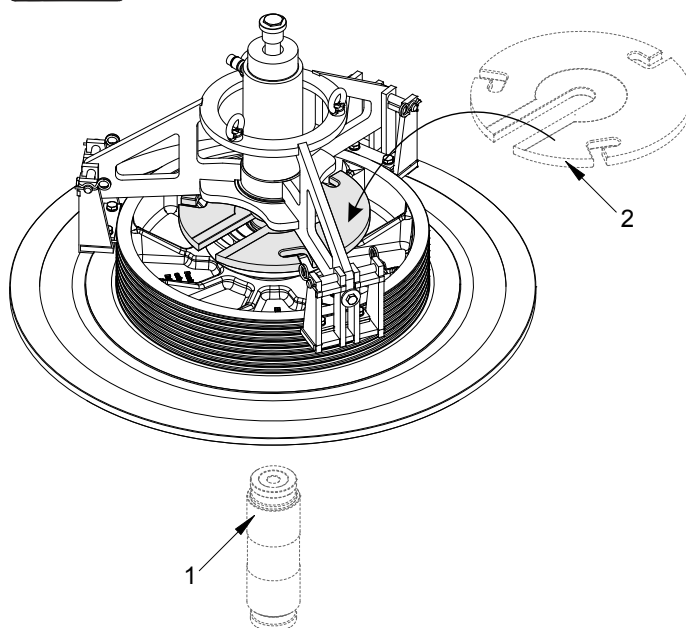
Related information

- [Use hydraulic jack with hand pump \(105\)](#)

a06759e9

5.3 Loosen top bearing

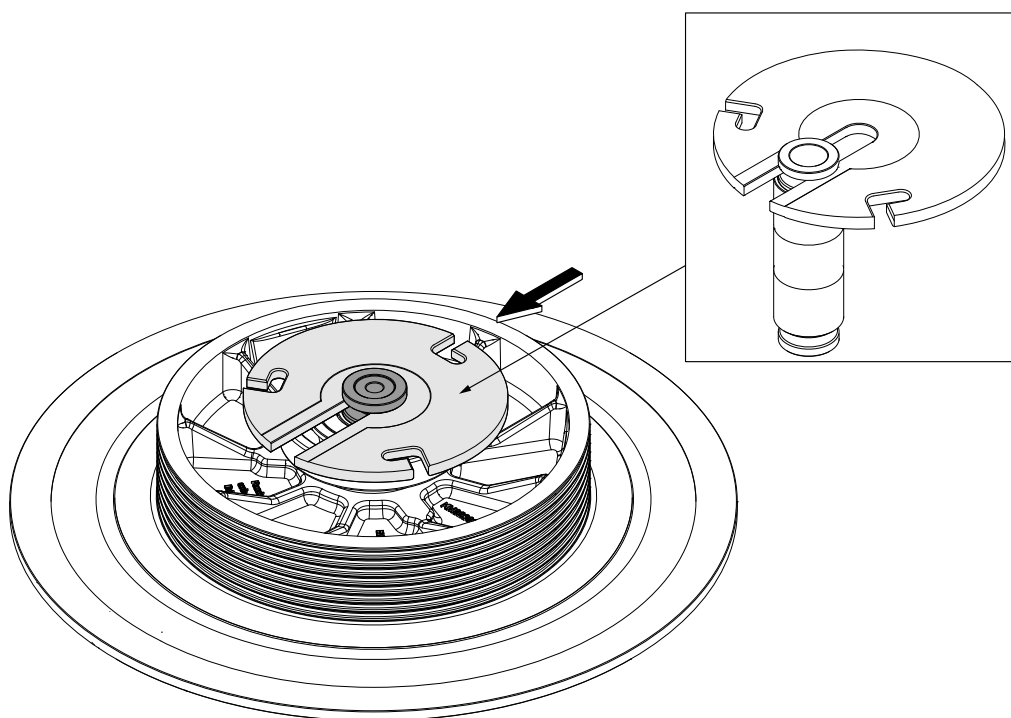
1. Install the top extractor tool bushing (1) inside the axle shaft. Place the bearing removal plate (2).



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2. Slide the bearing removal plate (2) in place to the bushing (1).

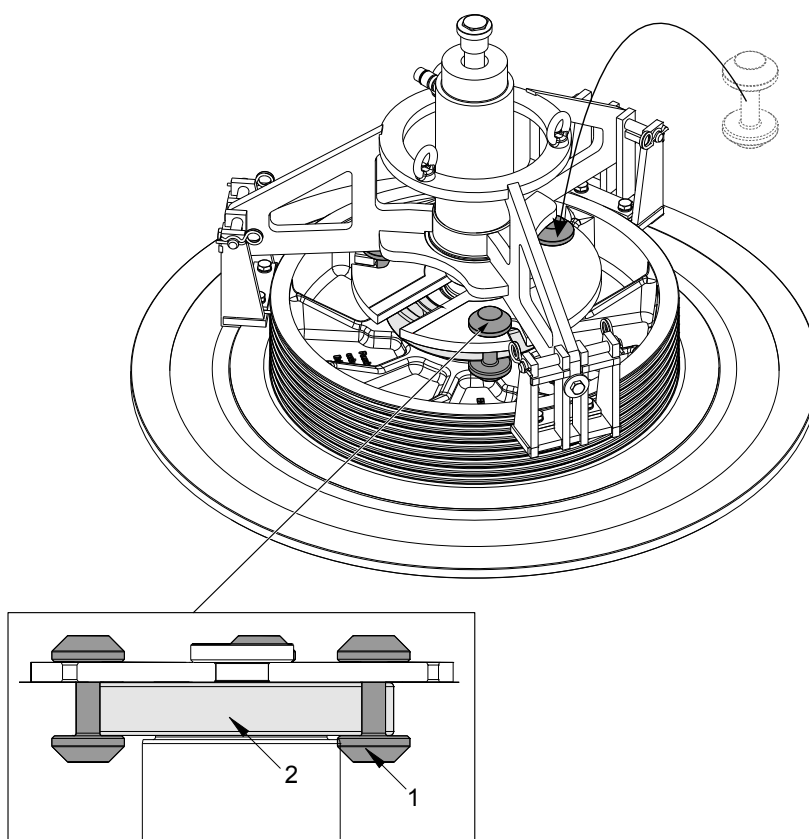




5ec46d93

3. Install the three bearing removal claws (1) to clamp the top bearing (2).

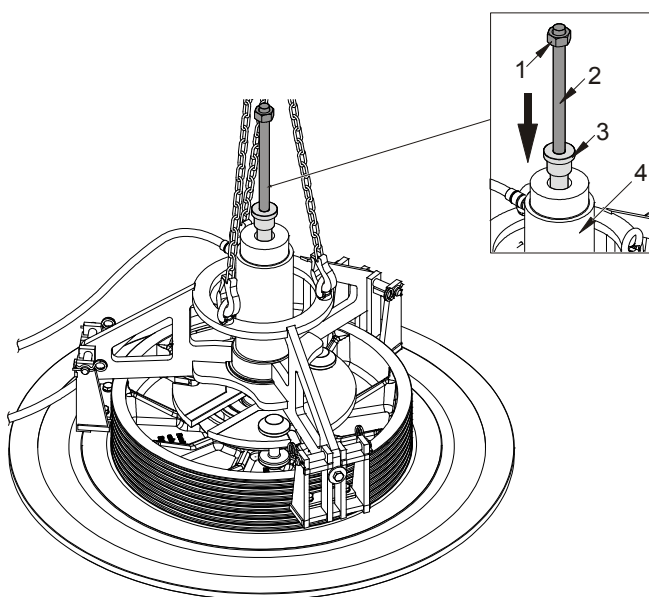




852c97b6

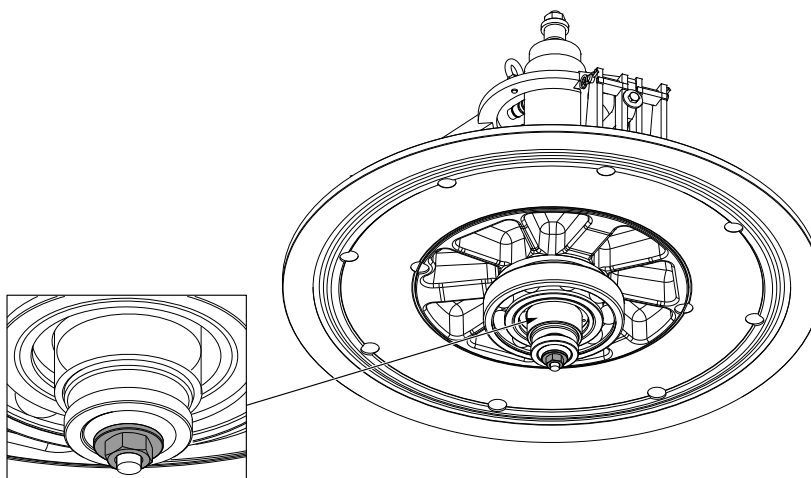
- Bearing removal claws (1) fit in between the ribs of the traction sheave.
4. Screw the top nut (1) to the end of the threaded bar (2). Install a washer (3) on top of the hydraulic jack (4). Insert the threaded bar (2) through the washer (3) into the hydraulic jack (4).





d1980447

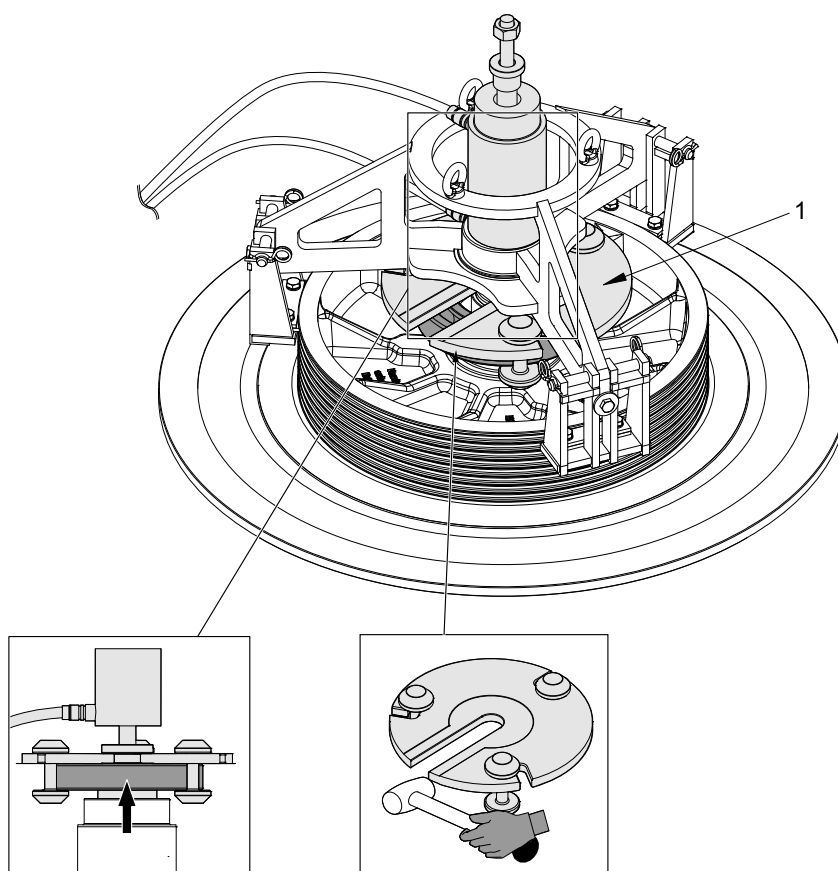
5. Install a washer and screw the bottom nut to the threaded bar.



86e74c15

6. Tighten the top nut to hand tight and pump the top bearing off the axle shaft.





65c879c6

Tap the side of the bearing removal plate (1) with a hammer to ease off the bearing if it seems to be stuck.

WARNING:

Always wear eye protection when using a hydraulic jack.

NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

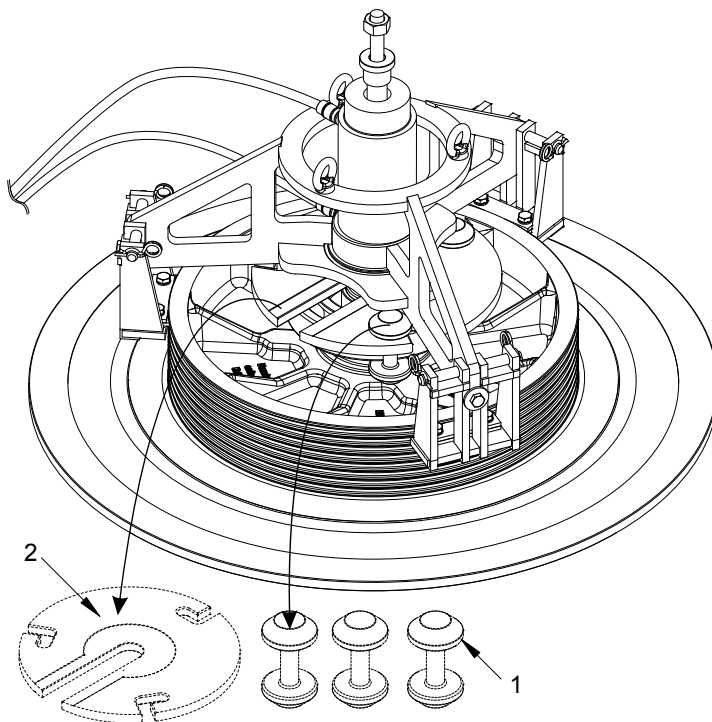
NOTE:

Once the pressure reaches 300 bar on the manometer, stop pumping.

NOTE:

If the bearing is stuck and the pump in use permits a pressure greater than 300 bar, pressure up to 500 bar at maximum to be used with caution.

7. Remove the bearing removal claws (1) and the bearing removal plate (2).



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NOTE:

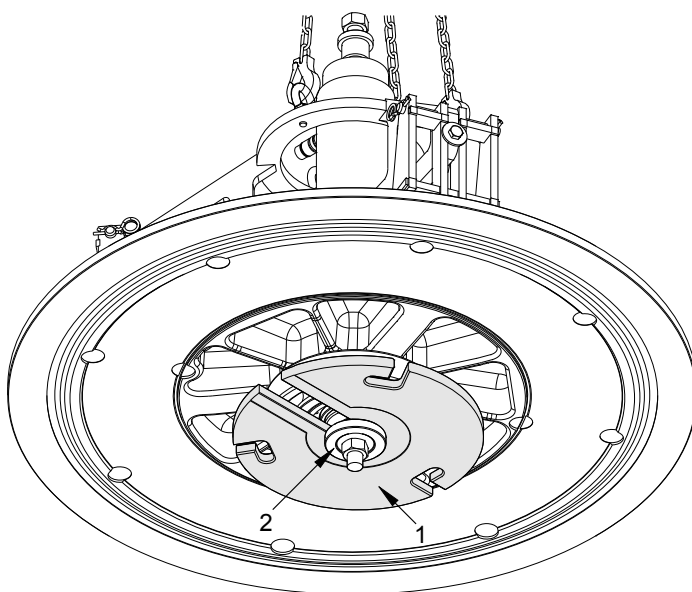
Check the bearing fit surface on the rotor. If the bearing fit surface is damaged, the rotor must be replaced. Do not proceed with lower bearing removal in this case.

be26fab6

5.4 Remove lower bearing

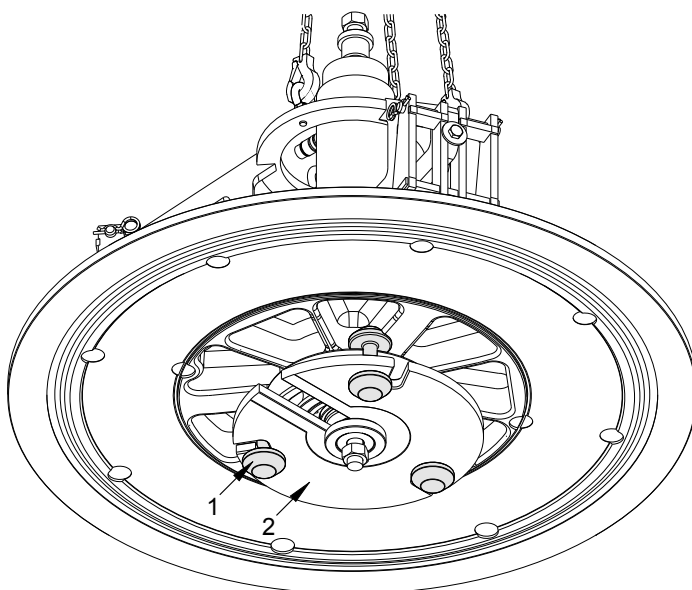
1. Slide the bearing removal plate (1) to the bushing (2).





7a253d51

2. Install bearing removal claws (1) to the bearing removal plate (2).



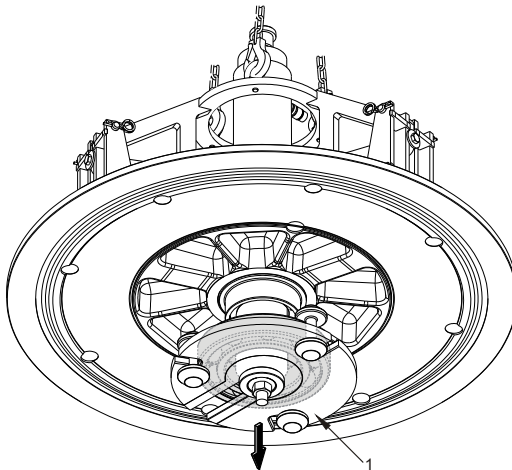
704728e0

NOTE:

Do not remove the threaded bar at this point.

3. Pump the lower bearing off.





eb3f18ac

Tap the side of the bearing removal plate (1) with a hammer to ease off the bearing if it seems to be stuck.

WARNING:

Always wear eye protection when using a hydraulic jack.

NOTE:

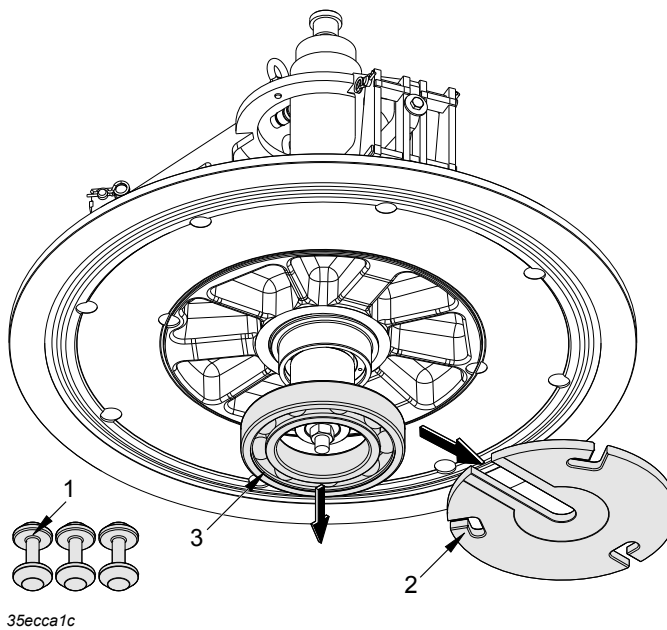
Once the pressure reaches 400 bar on the manometer, check the tool alignment and do not continue until everything is according to instruction. Pressure can increase then up to the limits specified on the hydraulic tools.

NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

4. Remove the bearing removal claws (1), the bearing removal plate (2) and the lower bearing (3).





35ecca1c

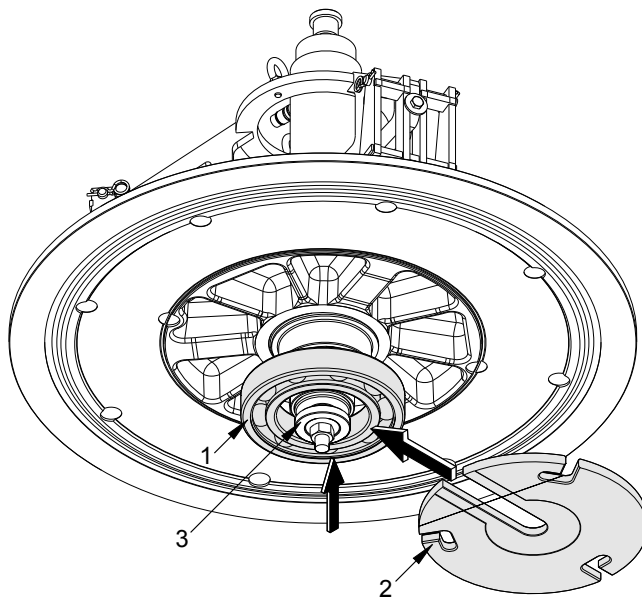
Clean the bearing fit surface with cleaning spray and cloth.

Check that the bearing fit surface is not damaged.

385f0835

5.5 Install lower bearing

1. Install the new lower bearing (1) and slide the bearing removal plate (2) to the bushing (3).

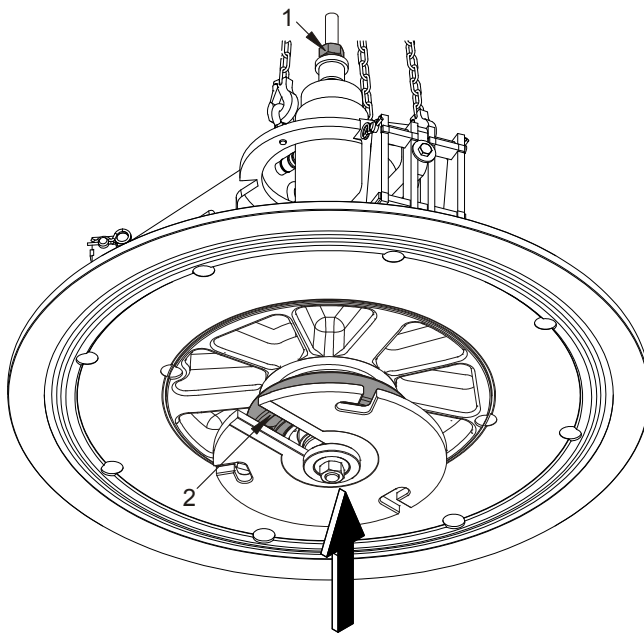


544c5bd0

CAUTION:

Do not wipe the new bearing surface if the bearing comes directly from the plastic package.

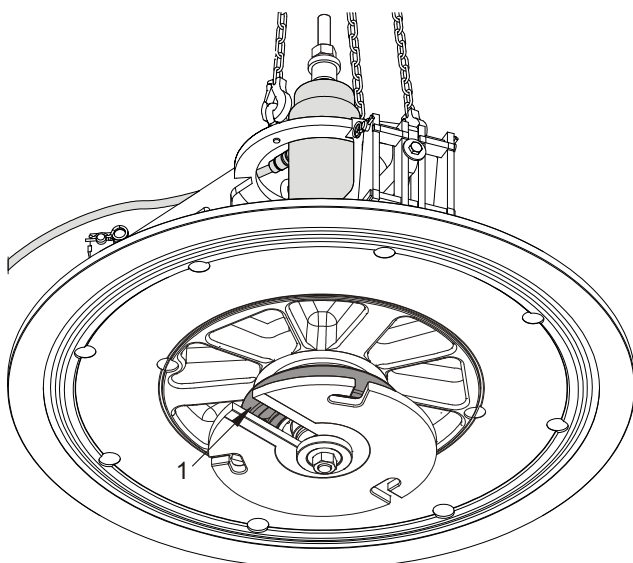
2. Tighten the top nut (1) so that the lower bearing (2) moves to the start of the bearing fit area in the axle shaft.



0fb2eb71

3. Pump pressure to pull the lower bearing (1) in place.
Continue pumping until the pressure reaches 300 bar on the manometer.





977df8a6

WARNING:

Always wear eye protection when using a hydraulic jack.

NOTE:

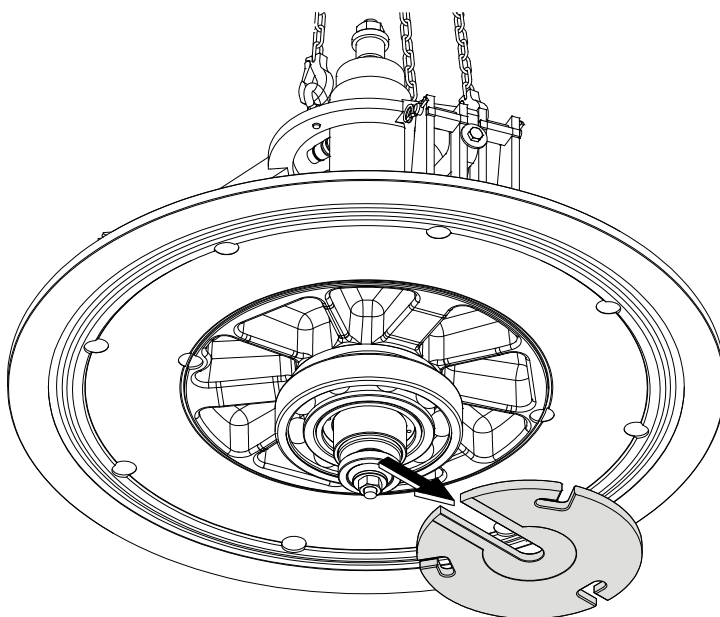
Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

NOTE:

Before starting pumping, check that the lower bearing is aligned evenly.

4. Remove the bearing removal plate.



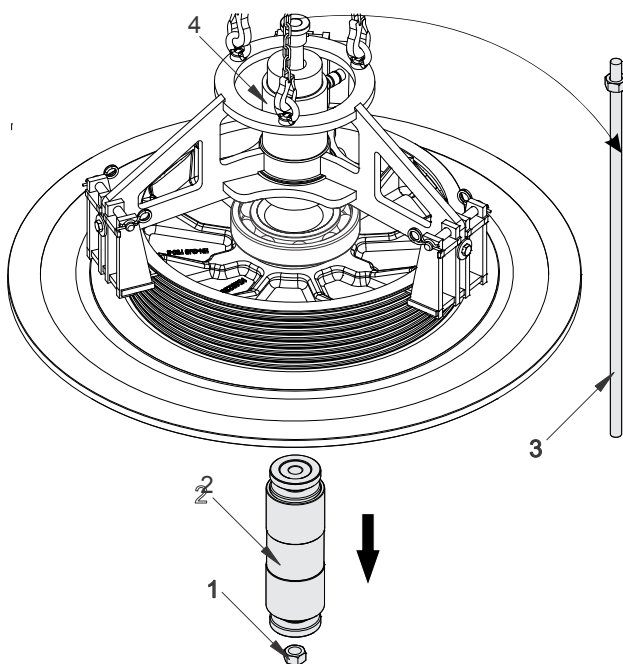


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5407dfb9

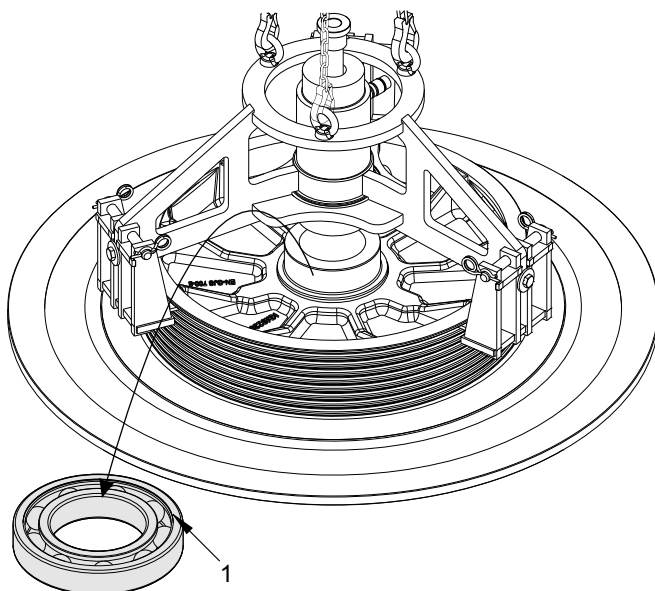
5.6 Replace top bearing

1. Remove the bottom nut (1) and the bushing (2). Remove the threaded bar (3) from the hydraulic jack (4).



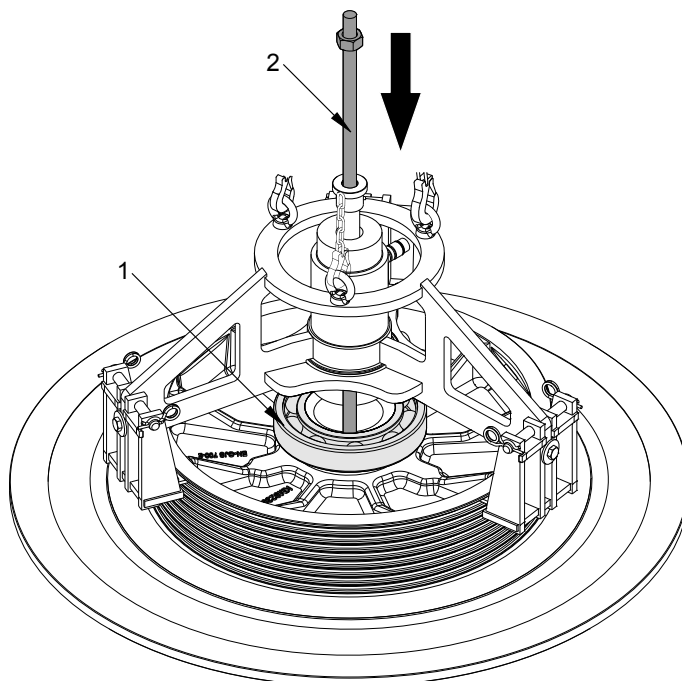
c8c6ae54

2. Remove the old top bearing (1). Clean the bearing fit surface with cleaning spray and cloth.



3b53459d

3. Install the new top bearing (1) and insert the threaded bar (2).

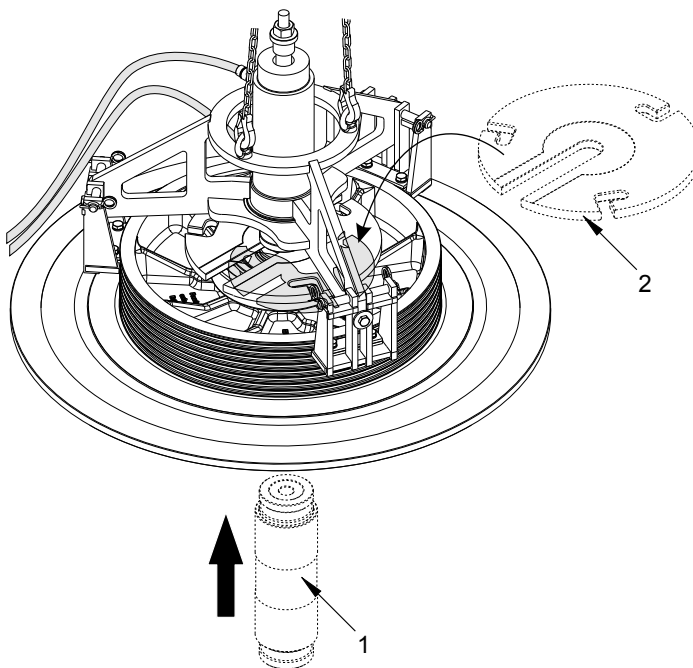


9979a79e

CAUTION:

Do not wipe the new bearing surface if the bearing comes directly from the plastic package.

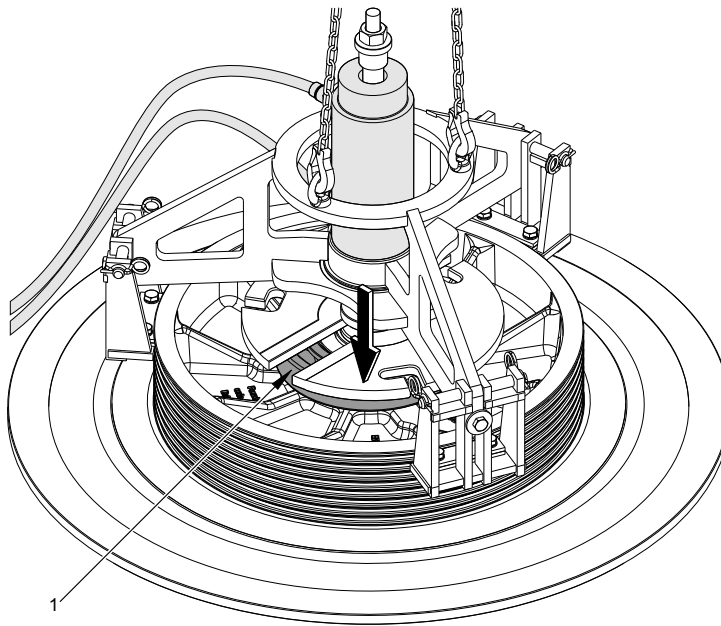
4. Install the bushing (1) to the axle shaft. Slide the bearing removal plate (2) to the bushing (1).



ad91e8e8

5. Pump pressure to push the top bearing (1) in place.





ad44f42d

- Continue pumping until the pressure reaches 300 bar on the manometer.

WARNING:

Always wear eye protection when using a hydraulic jack.

NOTE:

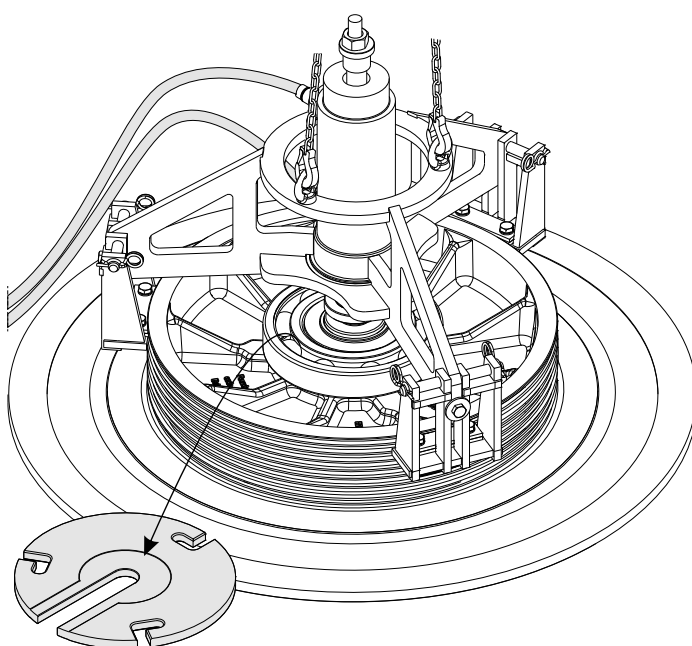
Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

NOTE:

Before starting pumping, check that the top bearing is aligned evenly.

6. Remove the bearing removal plate. Clean the stator assembly, rotor assembly and body1 with cleaning cloth.





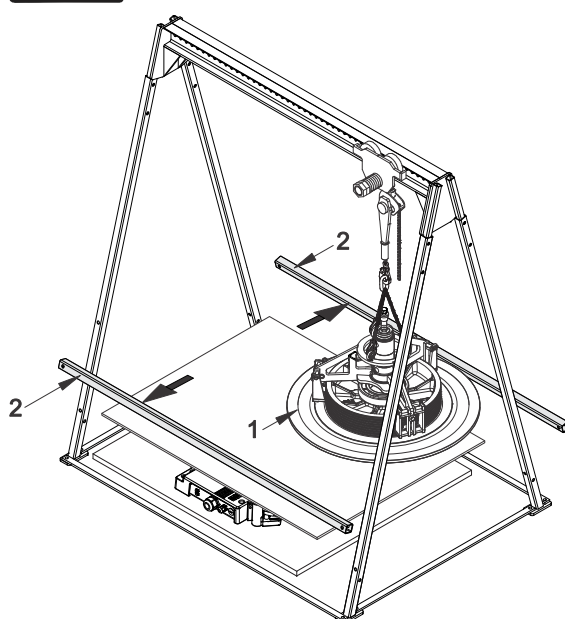
87a48855

7ba70ace

5.7 Assemble rotor

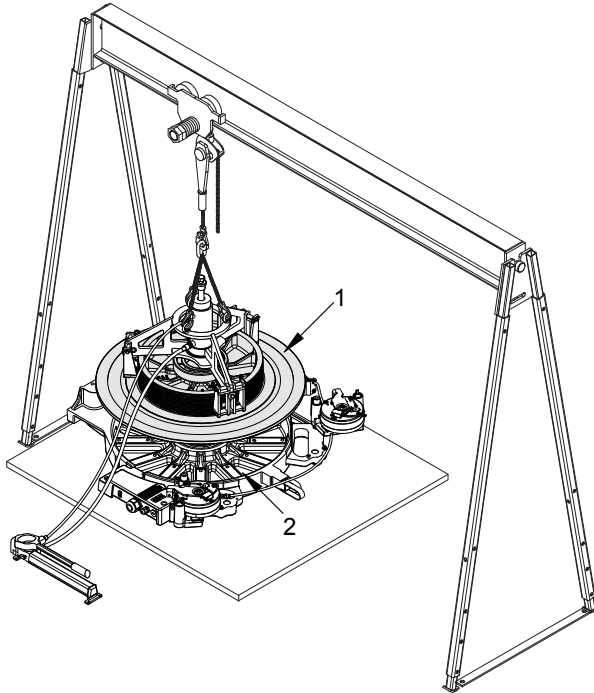
5.7.1 Assemble machine body

1. Lift the rotor assembly (1). Remove wood beams and the side beams (2) from the A-stand.



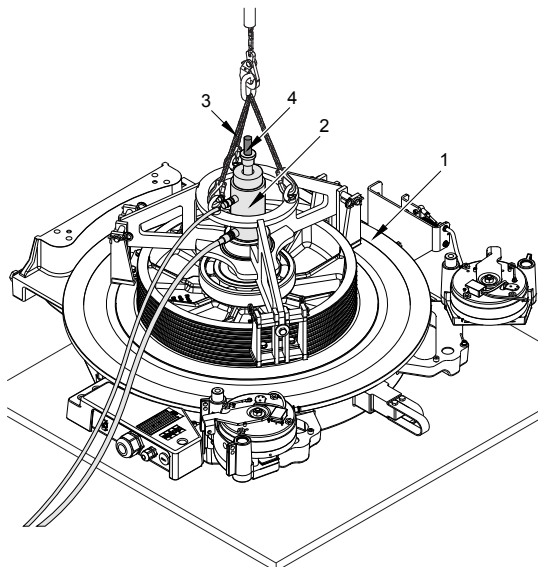
5e7260d6

2. Remove the plywood cover from the stator assembly (1). Move the rotor assembly (2) on top of the stator assembly (1).



814f9e6e

3. Pump the piston out of the hydraulic jack (2) and lower the rotor assembly (1) in place with the brake in the hand pump.



7641db3f

- Keep load chains (3) on while lowering with the hand pump.
- Gradually lower the rotor assembly (1) with both the chains and the pump.

NOTE:

Always wear eye protection when using a hydraulic jack.

NOTE:

For more information on using the hydraulic pump, refer to related information.

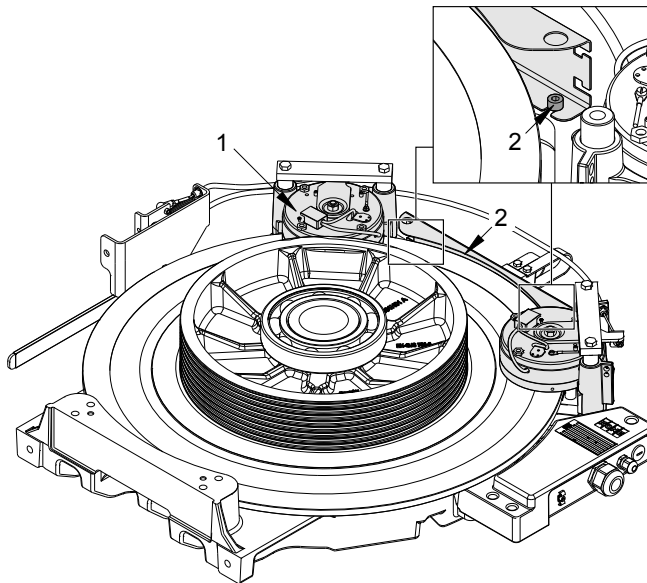
NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

NOTE:

Pull the rotor assembly (1) in place with threaded bar (4) and hydraulic jack (2) if needed.

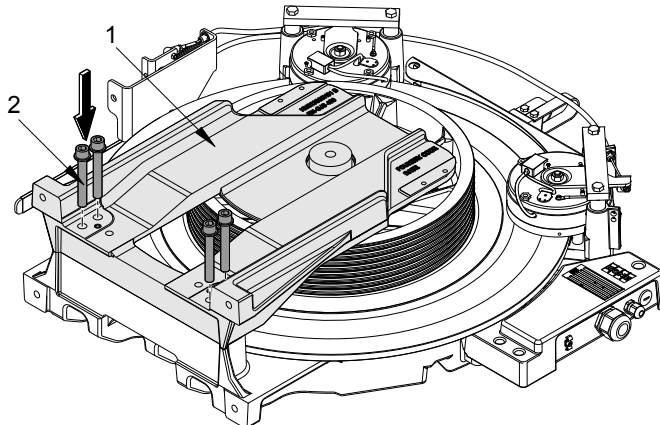
4. Install the brakes (1) in place, slide the top cover (2) back and tighten the screws.



57665a71

Refer to section *Installing the brake (From AS-04.08.034)*.

5. Install the body2 (1) with bolts (2).



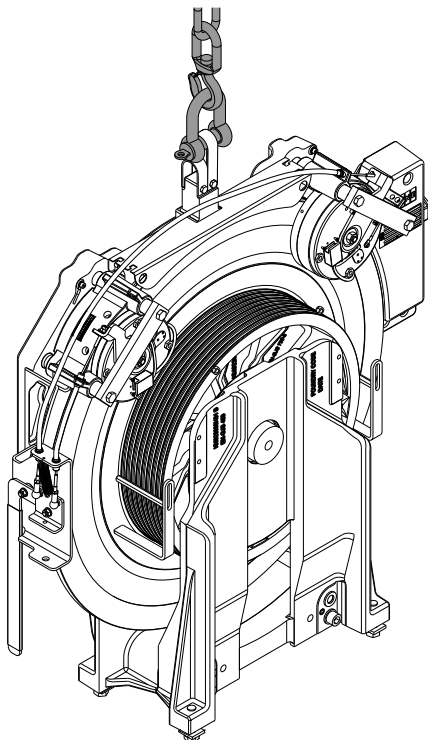
9960fdd0

- Check that body2 (1) attaches evenly at the bottom and to the bearing fit area.

CAUTION:

Do not knock the body 2 in place with a hammer. Check if there is any blocking object if it does not fit completely in place.

6. Lift the machine until it stands up.



b52274e9

NOTE:

Always keep the chain hoist attached when the machinery is in vertical position.

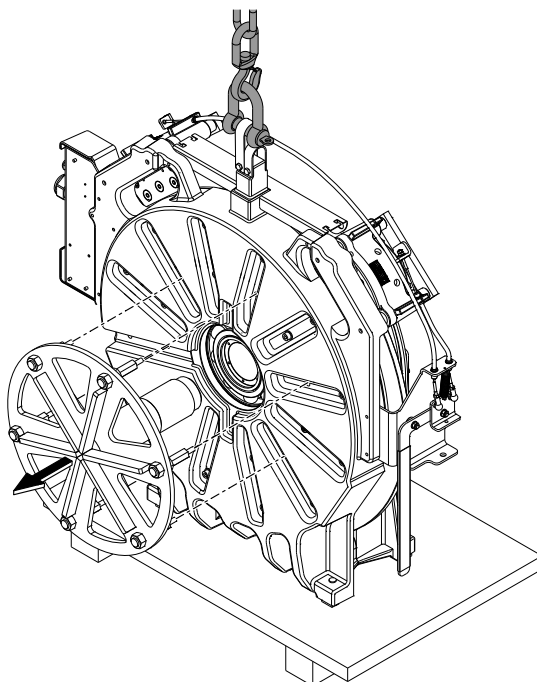
Related information

- [Install and adjust brake unit \(99\)](#)
- [Use hydraulic jack with hand pump \(105\)](#)

8a22d825

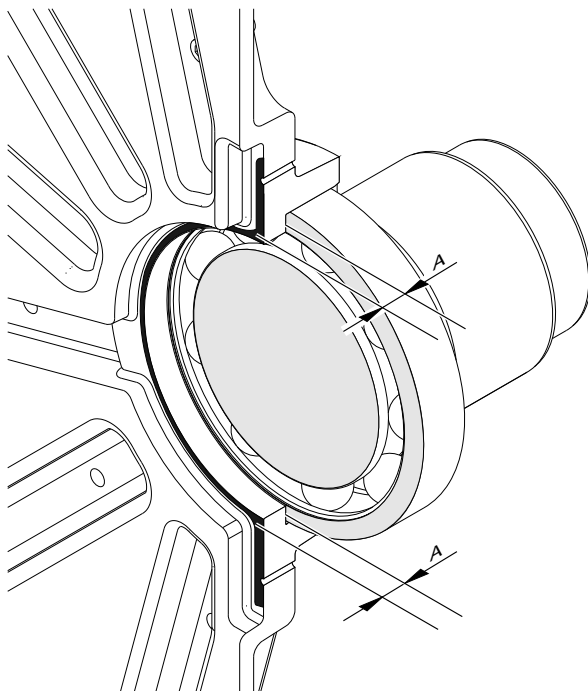
5.7.2 Check bearing position

1. Remove the lower extractor tool.



056463e9

2. Measure distance A.



13750223

- Check that the bearing surface is in line with the center axle top surface.
- Write down the upper and lower results, compare these to the results taken before bearing replacement.

NOTE:

Pay attention not to measure distance A from bearing shield.

NOTE:

If values differ remarkably, perform the following.

- Repeat steps from section *Install lower extractor tool* through *Install top extractor tool*.
- Install the hydraulic jack in place and pump to pull the assembly together.
- Pump until the pressure reaches 300 bar on the manometer.

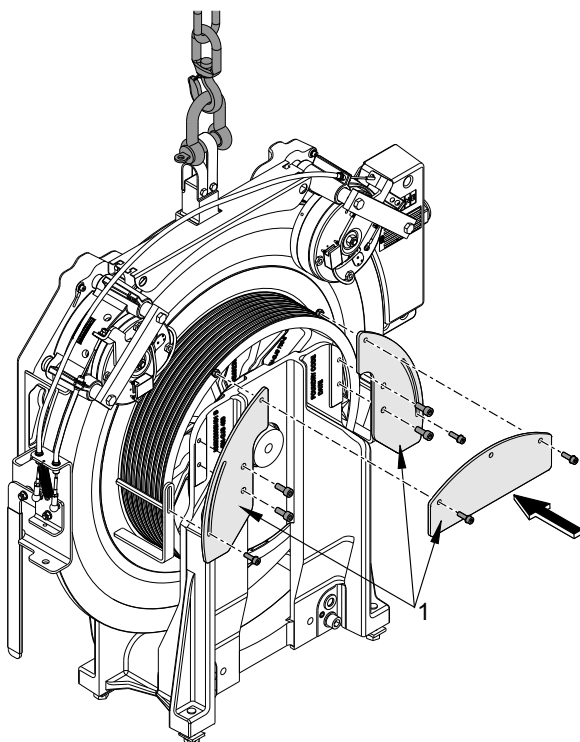
Related information

- [Install lower extractor tool \(34\)](#)
- [Install top extractor tool \(37\)](#)

5d861b3d

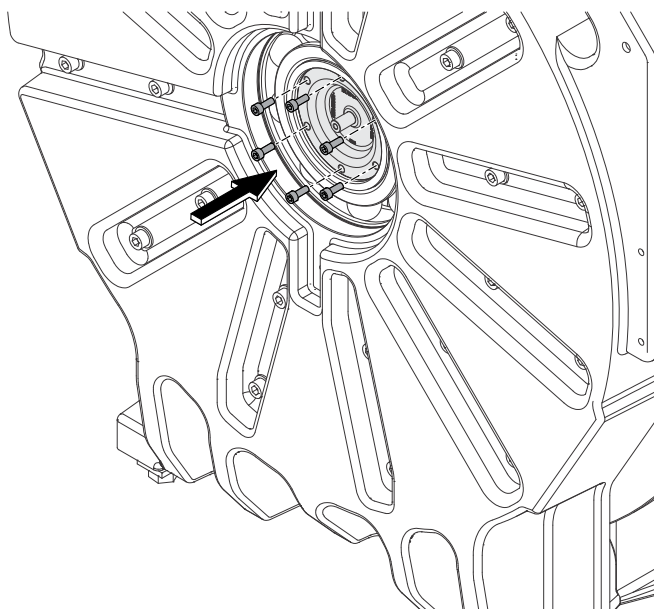
5.7.3 Install removed parts

1. Install the rope guards (1).



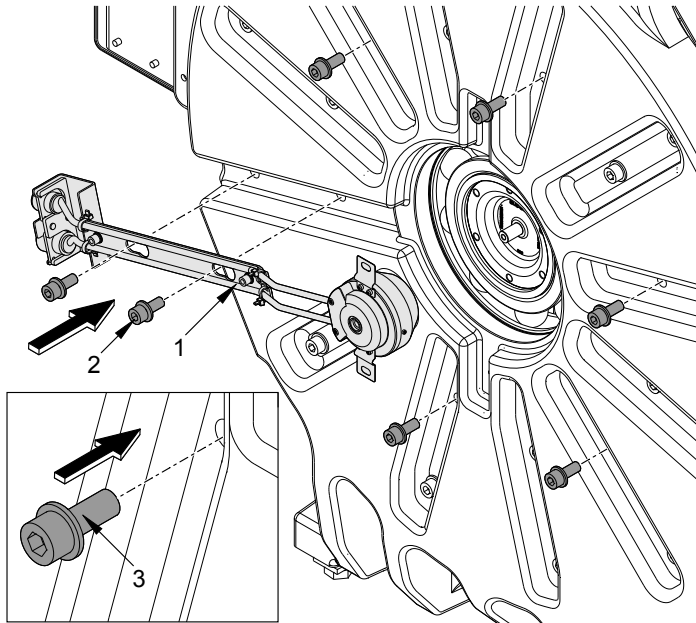
e6791005

2. Install the encoder shaft flange.



65c71c92

3. Install encoder wire holder plate (1) and the other hex socket screws (2, 3).



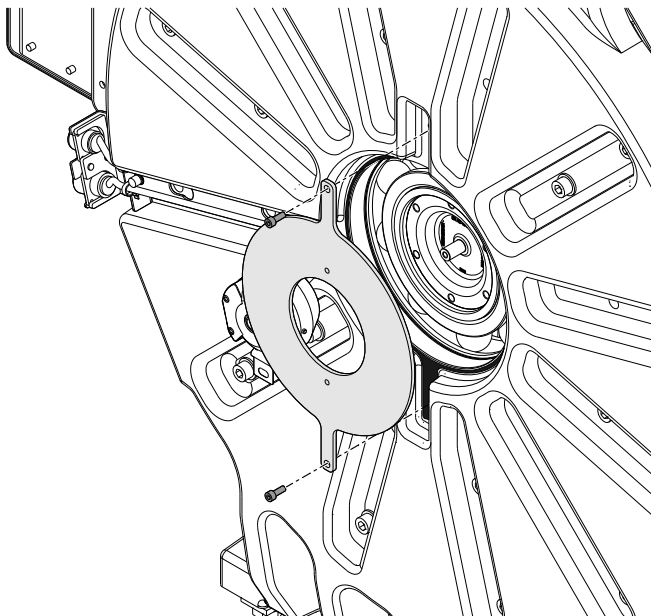
be1cff3d

- Replace the M10 × 25 screws if needed.
- Tighten the M10 × 25 screws to 49 Nm.

CAUTION:

Use only hex socket screws M10 × 25. Longer screws will damage the machine.

4. Apply new silicone glue between cast surface and the cover plate. Install the cover plate.

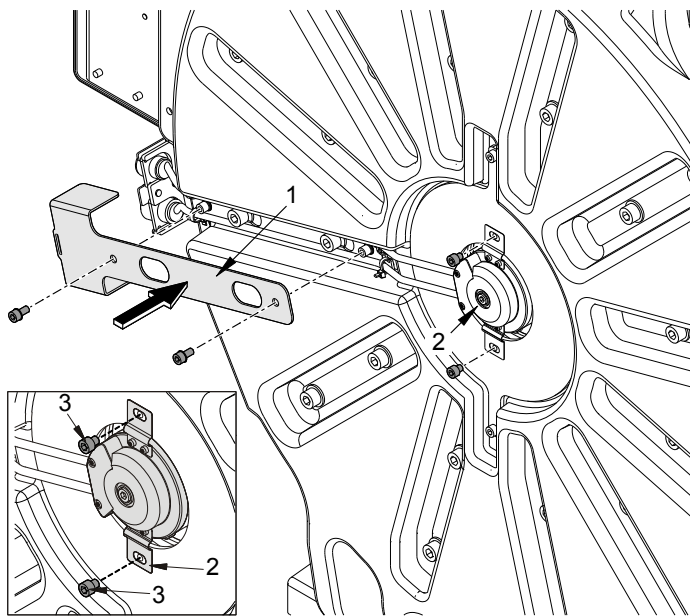


4e55a13b

NOTE:

Do not exceed the amount of glue used before. Check that all old glue is removed at first.

-
5. Install washers between the encoder wire cover plate (1) and the cast surface. Install the encoder (2) and the encoder wire cover plate (1).

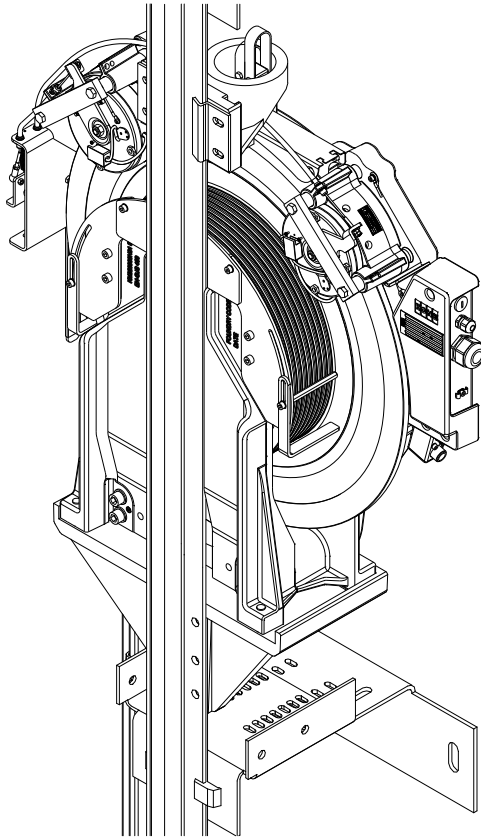


538059ce

NOTE:

If encoder fixing screws (3) are replaced, new thread locking compound must be added. Use Loctite 243 or equivalent.

6. Install the machine back to shaft / machine room.



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- For elevator with machine room, follow the methods described in AS-04.06.002.
- For machine-room-less elevator, follow the methods described in AS-04.06.001.
- Test the brakes according to section *One-sided electrical brake test with no load* (From ASG-04.06.001).

Related information

- [Perform one-sided electrical brake test with 0% load \(LCE\) \(103\)](#)
- [AS-04.06.001 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room-less Elevator](#)
- [AS-04.06.002 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room Elevator](#)

e76987dc

6 REPLACE ROTOR FOR MX14 MACHINE

6.1 Safety measures



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The following safety measures must be carried out before the actual work:

1. Protect the floor surface.

NOTE:

With rubber mat or dampers.

-
2. Secure the working area with safety fences.
 3. Make sure that all required equipment is on the work site.
 4. Install the A-stand.

NOTE:

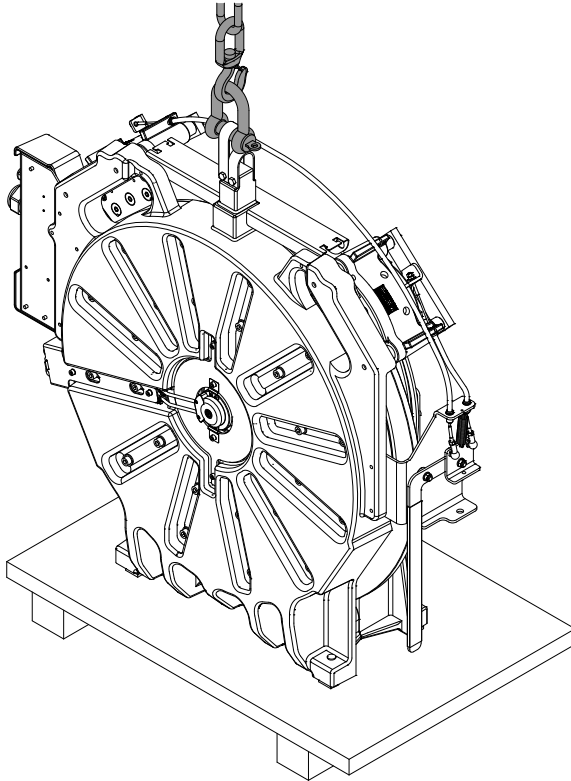
Fix the chain hoist to the A-stand.

-
5. Make sure that the A-stand modification is done correctly.

NOTE:

Do not attach side beams at this point.

6. Always keep the chain hoist attached when the machinery is in vertical position and when working with rotor assembly detached from the machinery.



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da123633

6.2 Remove rotor



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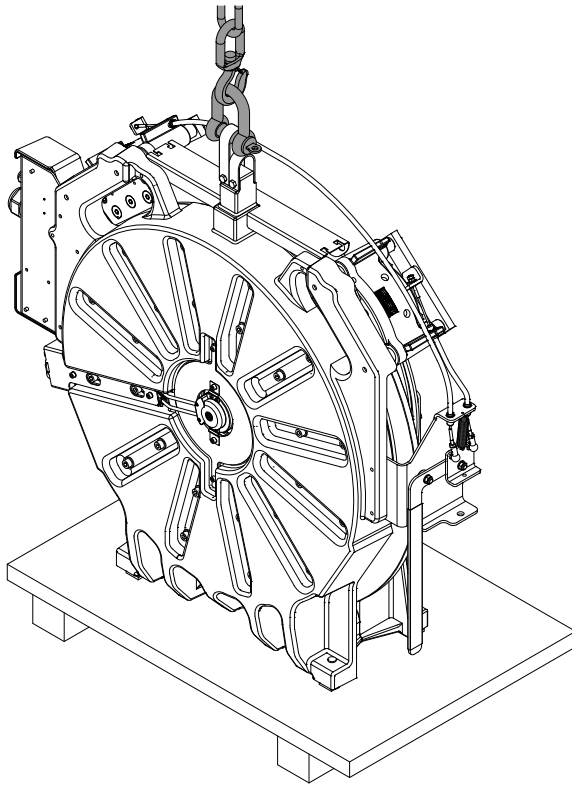
1. Remove MX14 machine from machine room / elevator shaft.
 - For more information on elevator with machine room, refer to the related information.
 - For more information on machine-room-less elevator, refer to the related information.

NOTE:

Always keep the chain hoist attached when the machinery is in vertical position.

NOTE:

Technician must make sure the tightness of the chain conditions before performing component replacement activity.

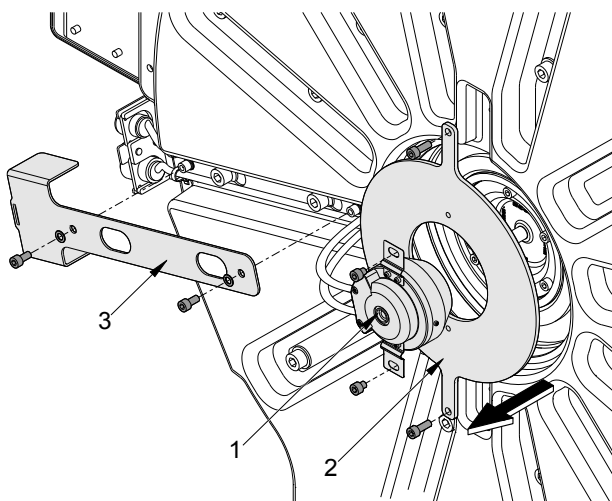


ff6f2b76a

2. Detach the encoder (1). Remove the cover plate (2) and the encoder wire cover plate (3).

NOTE:

Check that all the threads on the removed screws are intact. If not, check the threads on the machine. If machine threads are damaged, the whole machine must be replaced.



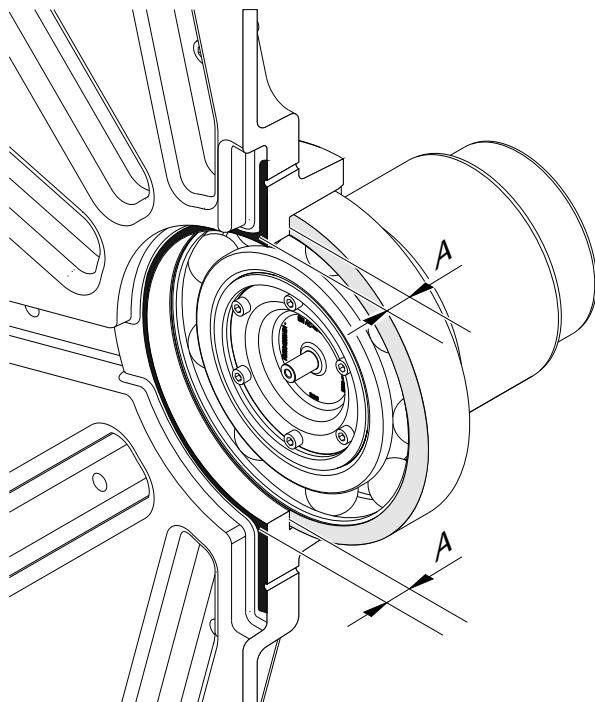
ff0ab1fb

3. Remove all black silicone residues from the cast surface. Measure distance A.
 - Check that the bearing surface is in line with the center axle top surface.

- Write down the upper and lower results for later replacement of the bearing.

NOTE:

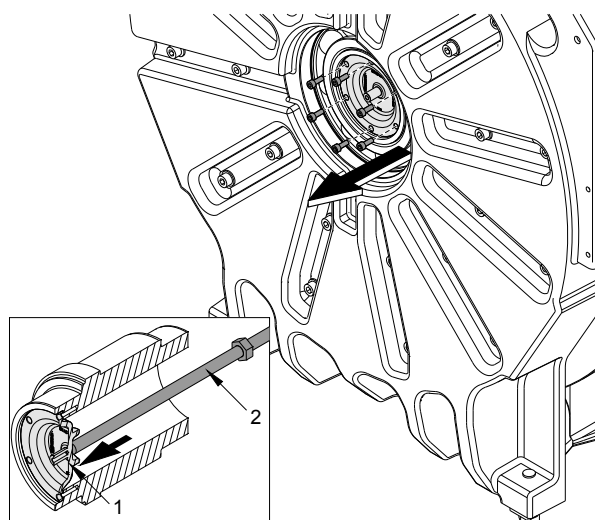
Pay attention not to measure distance A, from bearing shield.



d5ee209f

4. Remove the encoder flange (1).

- Use the threaded bar (2) for example.
- Insert the threaded bar (2) from body2 and tap the end with a hammer to loosen the flange (1).



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5. Check that the extractor tool bushings fit the axle elevator shaft.

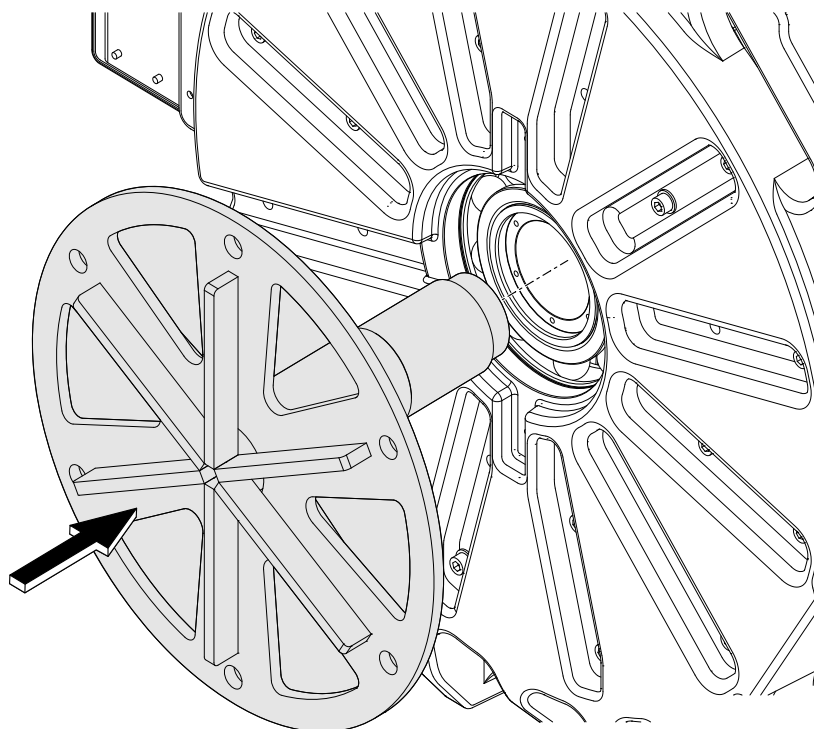
NOTE:

The whole machine must be replaced if the bushings do not fit the axle elevator shaft.

NOTE:

Two persons must lift and install the tool.

Active and proper communication to be maintained between the two technicians.

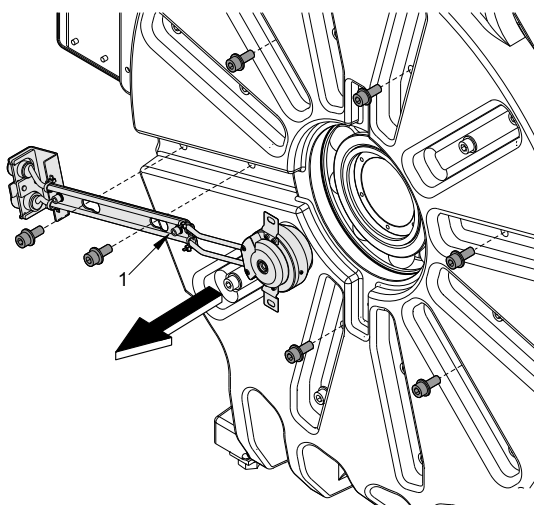


44c162ff

6. Remove the encoder.
- Remove the wire holder plate screws (1) and hex socket screws. (Counting from the encoded wire plate screws, every other hex socket screw should be removed.)
-

NOTE:

Check that all the threads on the removed screws are intact. If not, check the threads on the machine. If machine threads are damaged, the whole machine must be replaced.

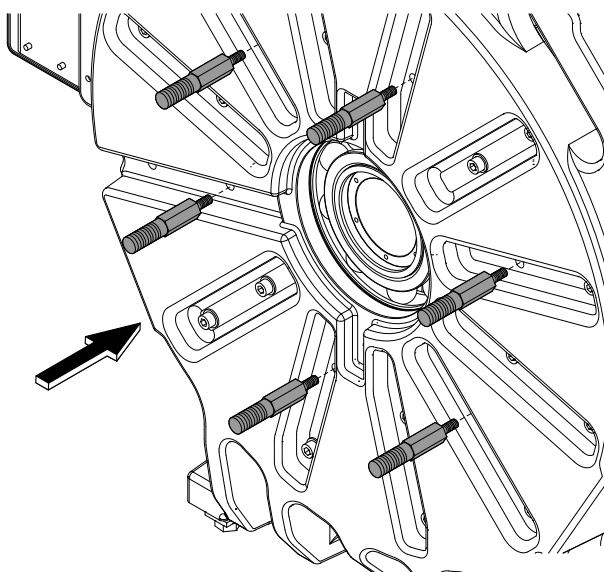


0c60e248

7. Install the six threaded bars to the open threaded holes.

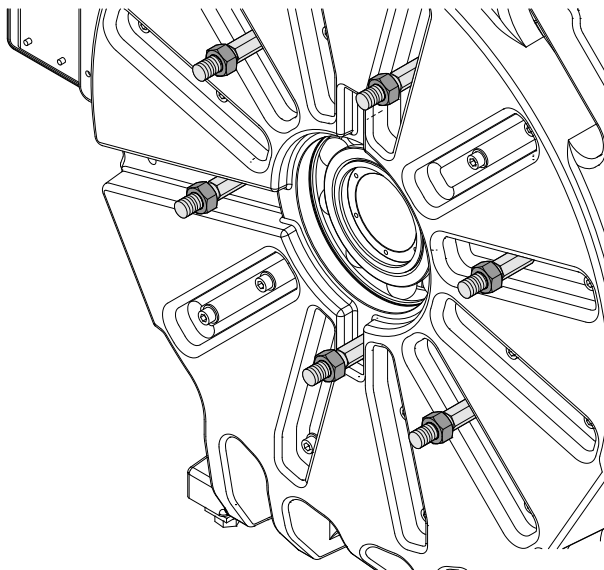
NOTE:

If removed screws are damaged, do not insert threaded bars.



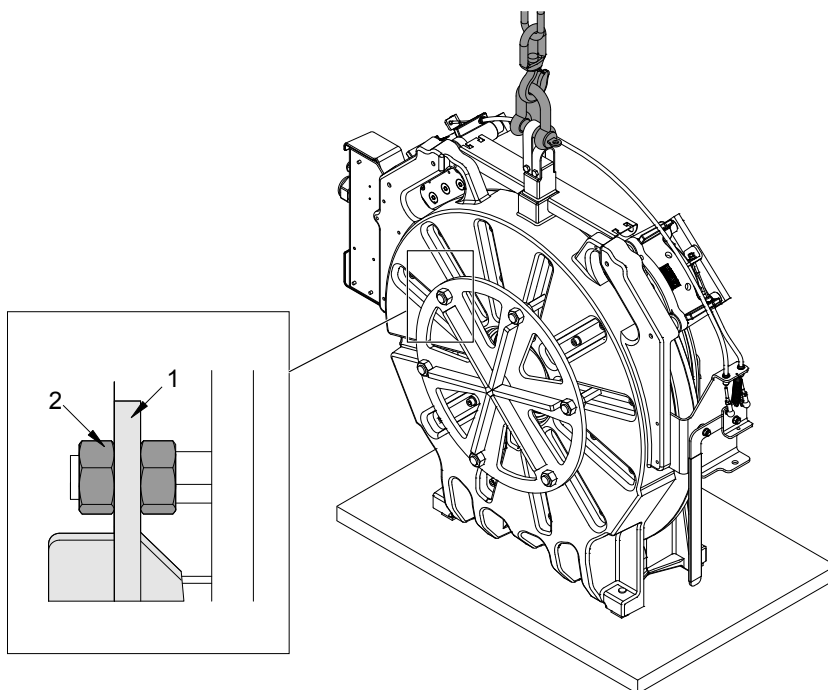
bd5b11f0

8. Screw a nut to each of the six threaded bars.



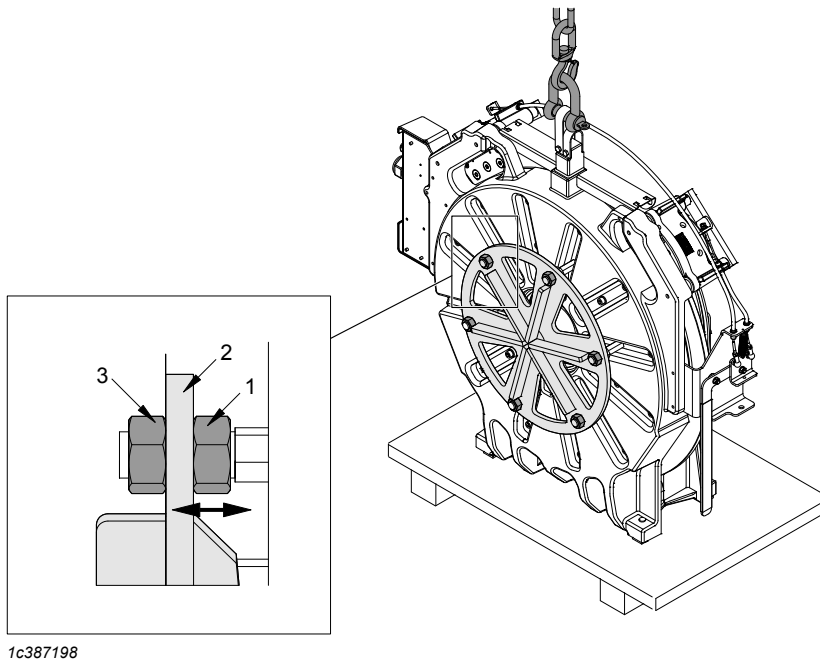
36607be4

9. Attach the lower extractor tool (1) to the six threaded bars and screw an outer nut (2) to each of the threaded bars.

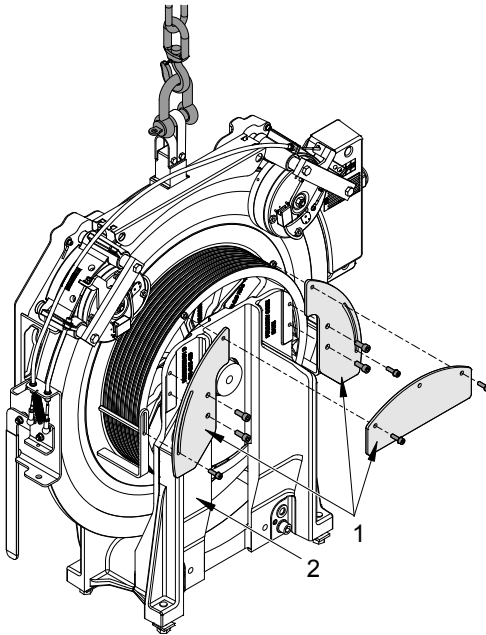


48440fd5

10. Orient the lower extractor tool (2) with the nuts (1). Tighten the outer nuts (3) with a wrench.



11. Remove the rope guards (1) from body2 (2).



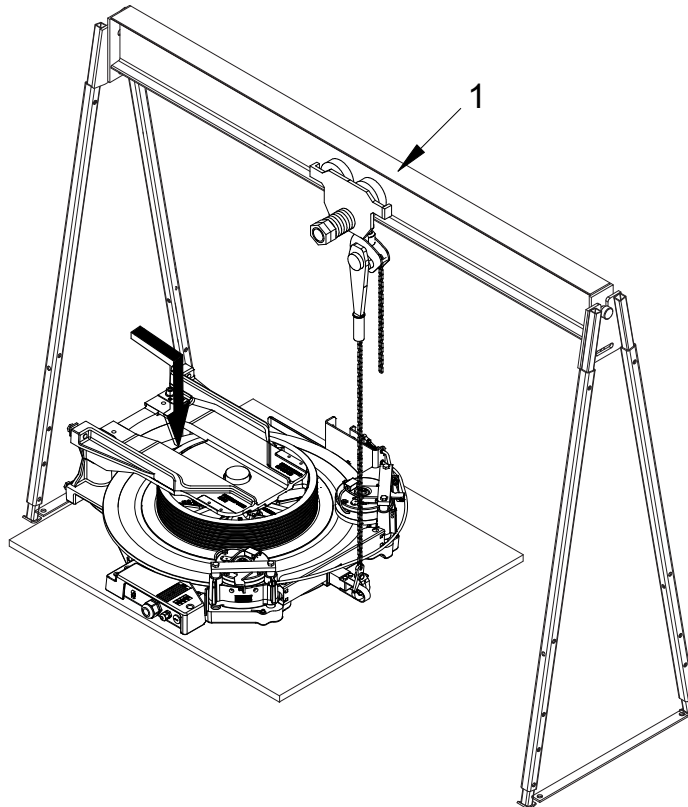
12. Use A-stand (1) and chain hoist (2) to lift and tilt the machine until it is laid on a plywood piece or rubber mat / pallet.
- With body2 facing up and body1 facing down.
 - With the installed lower extractor tool between the plywood piece or rubber mat / pallet and body1.

NOTE:

Place the machine on one end of the A-stand (1), not in the middle of it.

NOTE:

Two persons should be lowering the machine downwards initially by pushing the machine for this activity by supporting with chain hoist.



8a3326a6

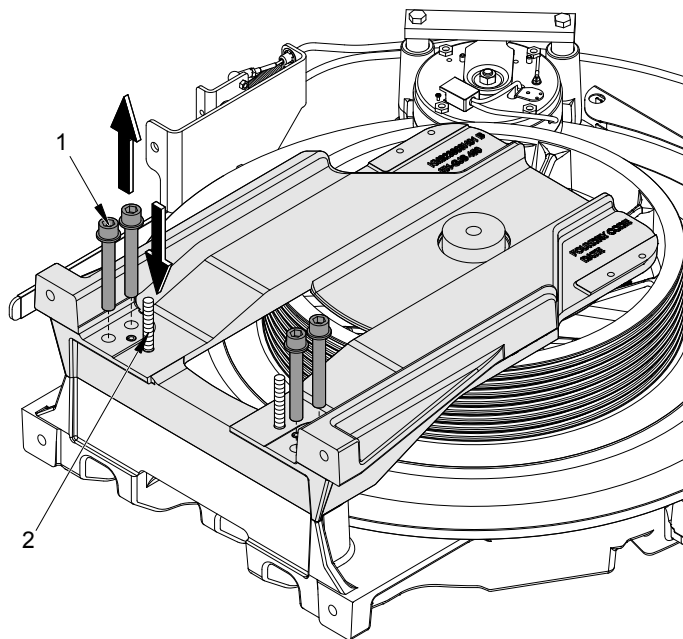
13. Remove the four screws (1) on body2 (two screws on each side). Insert two bolts (2) into the two threaded holes.
-

NOTE:

Before inserting bolts (2), check that there is a chamfer at the start of each threaded hole.

NOTE:

Use well-greased bolts.



5a70f5d0

14. Use an aluminum pry bar (1) to remove the body2.

- Use pry bar (1) and push out the other end of body2 by tightening bolts (2) in turn so that body2 is pushed upwards.

CAUTION:

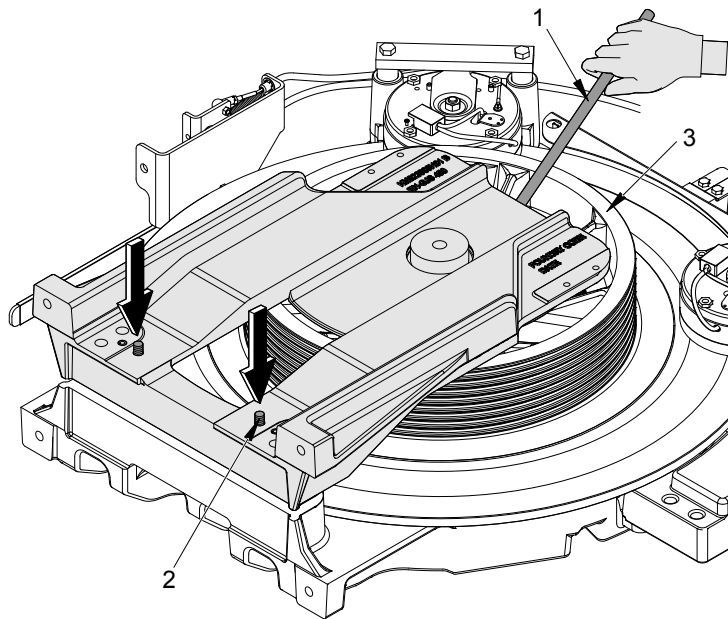
Use the aluminum pry bar (1) with care to avoid damaging the traction sheave (3).

NOTE:

Two persons should be involved in this activity.

One technician lifts the iron cross using a pry bar and another technician uses the sling belt to lift on other end.

Active and proper communication to be maintained between the two technicians.



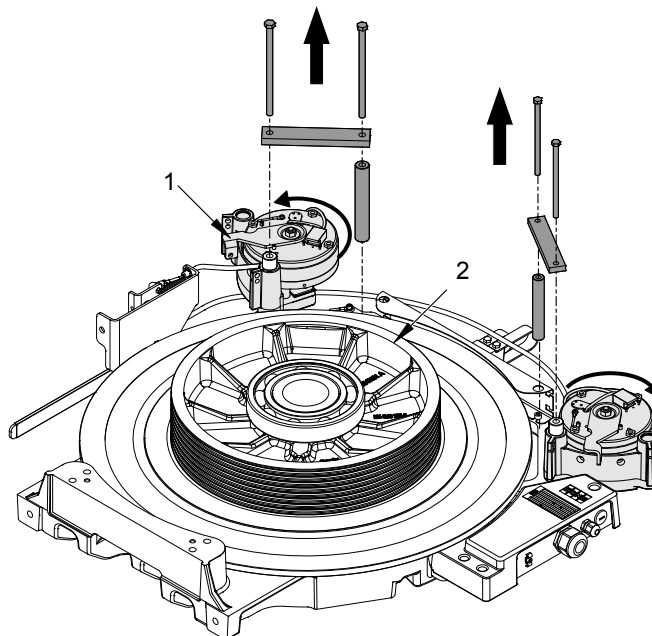
9f144981

15. Move brakes (1) aside from traction sheave (2).

For more information, refer to the related information.

NOTE:

Do not lift the brakes (1) away. Only turn it aside of traction sheave (2).

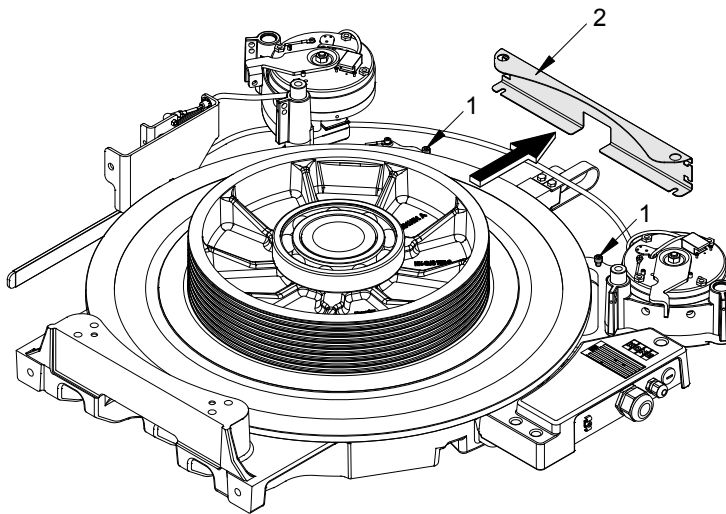


60b93f9c

16. Loosen the cover plate screws (1) and slide off the cover plate (2).

NOTE:

Do not remove the screws (1) completely.



28309267

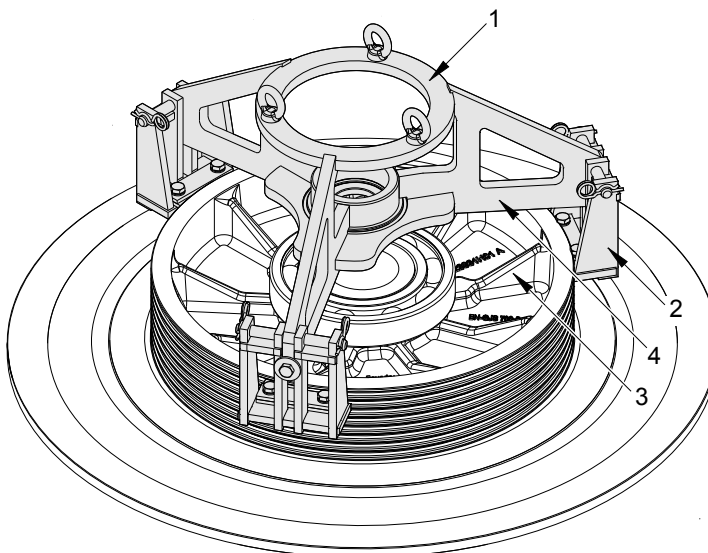
17. Install the top extractor tool (1) to the traction sheave grooves.

- Use the upmost groove to attach claws (2).
- Adjust the top extractor tool (1) so that the traction sheave ribs (3) are aligned with the wings (4) of the tool.

NOTE:

Two persons must lift and install the tool.

Active and proper communication to be maintained between the two technicians.



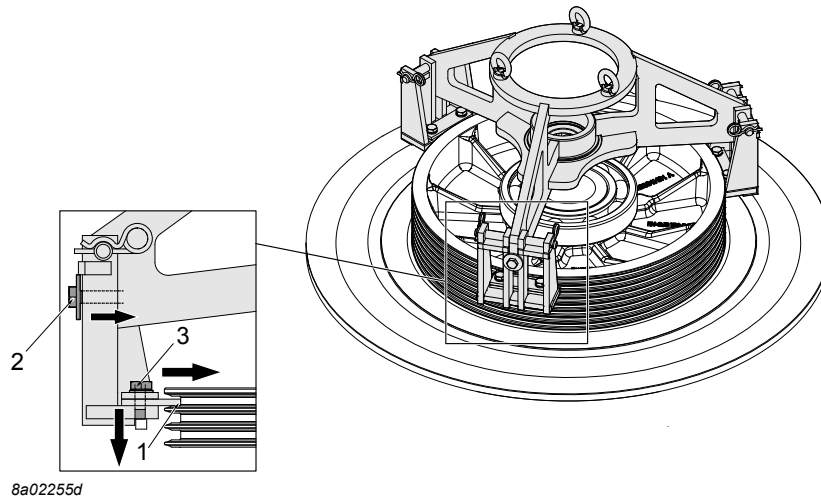
d393b80a

18. Check that the claws (1) reach the bottom of the groove undercut. Tighten the claw holder screws (2).

- Adjust claw depth if needed by screws (3).

NOTE:

Tighten the claw holder screws (2) evenly so that the tool is centered with the traction sheave.



19. Install the hydraulic jack (1) and connect the hand pump (2).
For more information on using the hydraulic pump, refer to the related information.
-

WARNING:

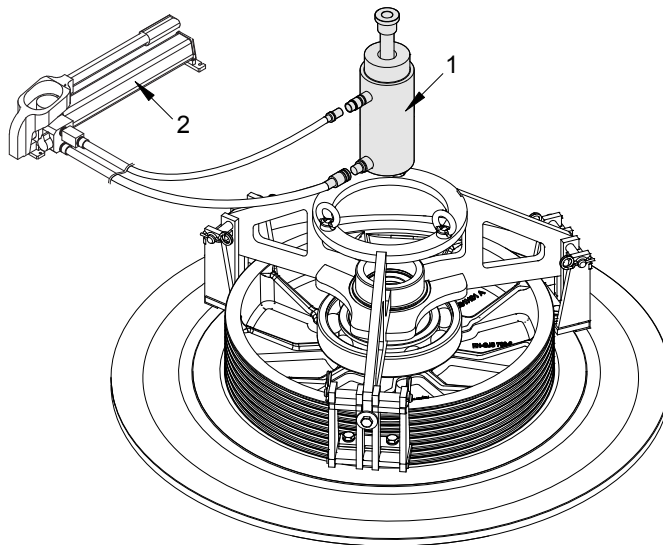
Always wear eye protection when using a hydraulic jack.

WARNING:

The rotor has a strong magnetic field and a strong attraction force. Handle the rotor carefully and take care not to get your hands crushed between the magnet and the tools.

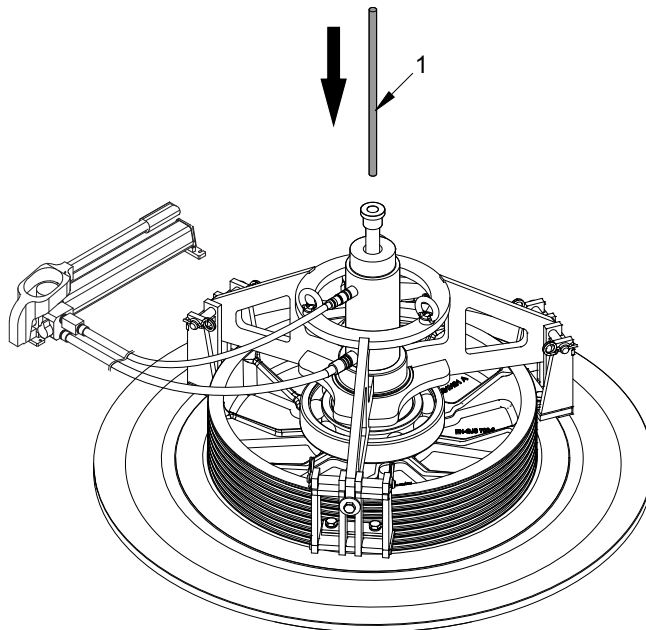
NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.



5305bb91

20. Check tool alignment by inserting the M20 threaded bar (1).
– Adjust if necessary.



e8939c97

21. Pump pressure until rotor assembly (1) becomes loose. Lift the rotor assembly (1) while pumping it up.

WARNING:

Always wear eye protection when using a hydraulic jack.

CAUTION:

Before pumping, always check that the piston has enough room to move for the operation to be done. Constantly look at the piston to make sure it does not collide unintentionally while pumping.

NOTE:

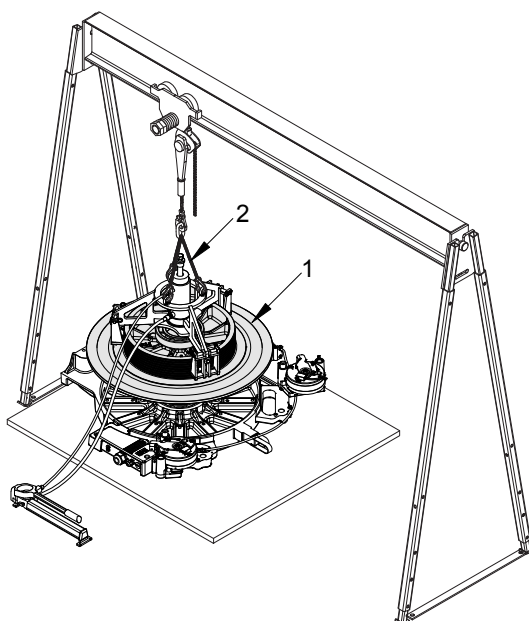
Once the pressure reaches 300 bar on the manometer, stop pumping immediately.

NOTE:

Adjust the 3-branch load chain (2) so that the lifted rotor assembly (1) does not exceed the height where the side beams can be attached.

NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.



da3cd135

22. Lift and move rotor assembly (1) to the other end of the A-stand. Place plywood cover (2) on top of stator assembly (3) to protect it against dropping objects.
-

WARNING:

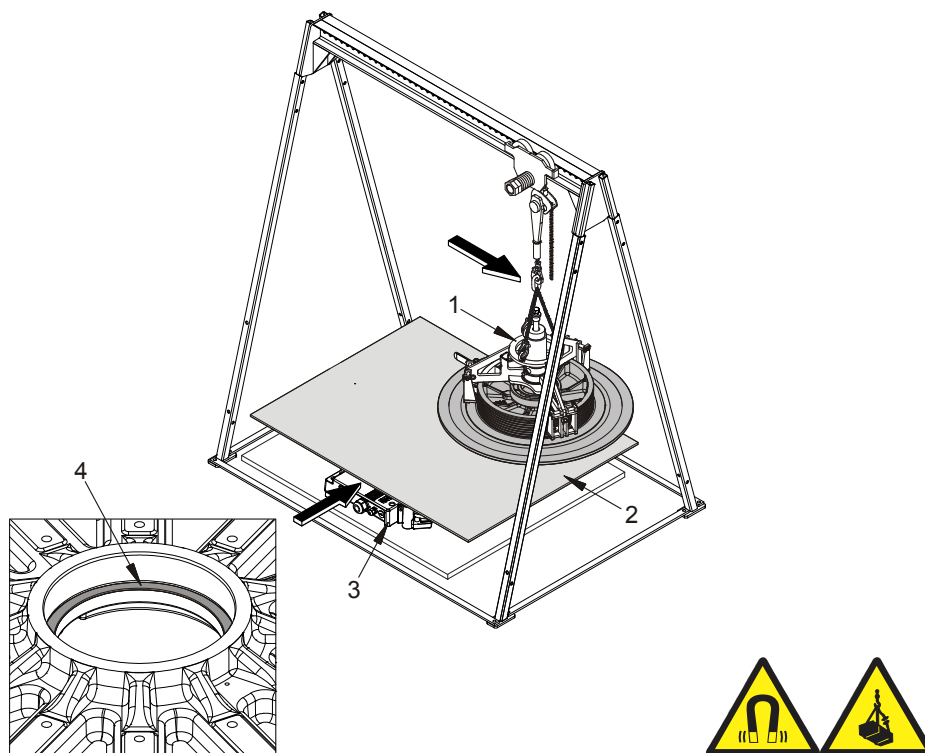
Working under a suspended load is not allowed. Only replace bearings when chains are attached, and beams are installed under the rotor assembly.

NOTE:

Rotor assembly with lifting tools weighs approximately 200 kg. Consider weight when lifting.

NOTE:

Do not remove shim plates (4). If shim plates (4) stick to rotor assembly, return shim plates (4) to stator assembly.



23. Remove the old rotor.

Related information

- [AS-04.06.001 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room-less Elevator](#)
- [AS-04.06.002 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room Elevator](#)
- [Remove brake unit \(96\)](#)
- [Use hydraulic jack with hand pump \(105\)](#)

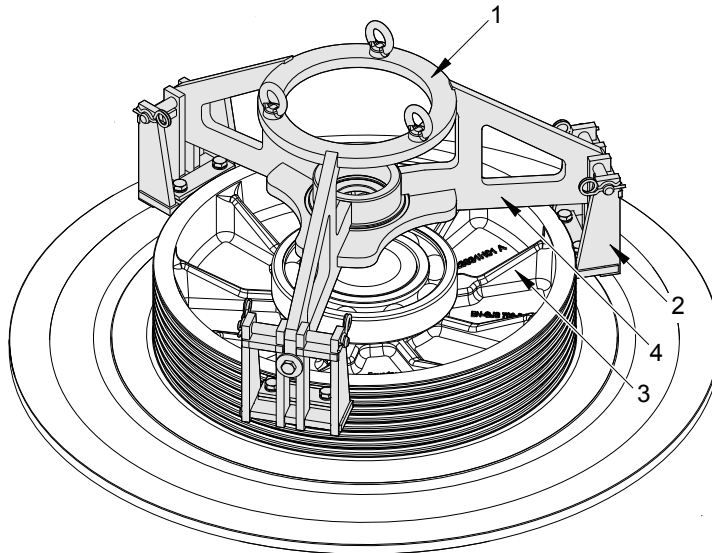
e47f1641

6.3 Install new rotor



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1. Unpack the new rotor and place it on the landing.
2. Install the top extractor tool (1) to the traction sheave grooves.
 - Use the upmost groove to attach claws (2).
 - Adjust the top extractor tool (1) so that the traction sheave ribs (3) are aligned with the wings (4) of the tool.

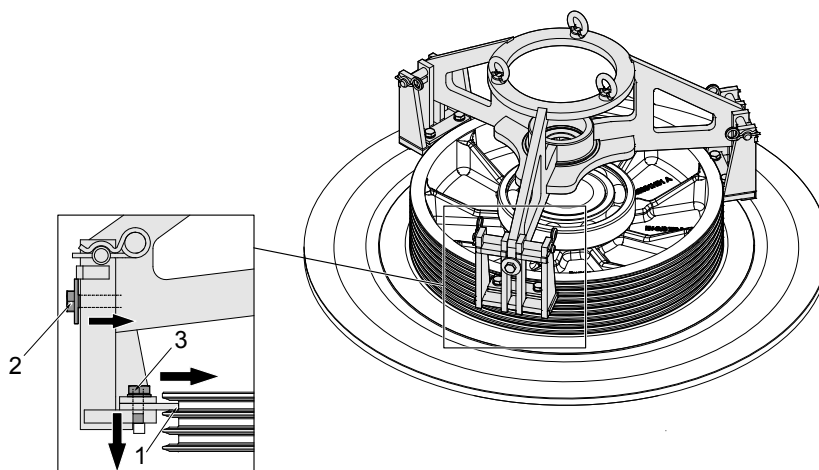


d393b80a

3. Check that the claws (1) reach the bottom of the groove undercut. Tighten the claw holder screws (2).
 - Adjust claw depth if needed by screws (3).

NOTE:

Tighten the claw holder screws (2) evenly so that the tool is centered with the traction sheave.



8a02255d

4. Install the hydraulic jack (1) and connect the hand pump (2).

For more information on using the hydraulic pump, refer to related information.

WARNING:

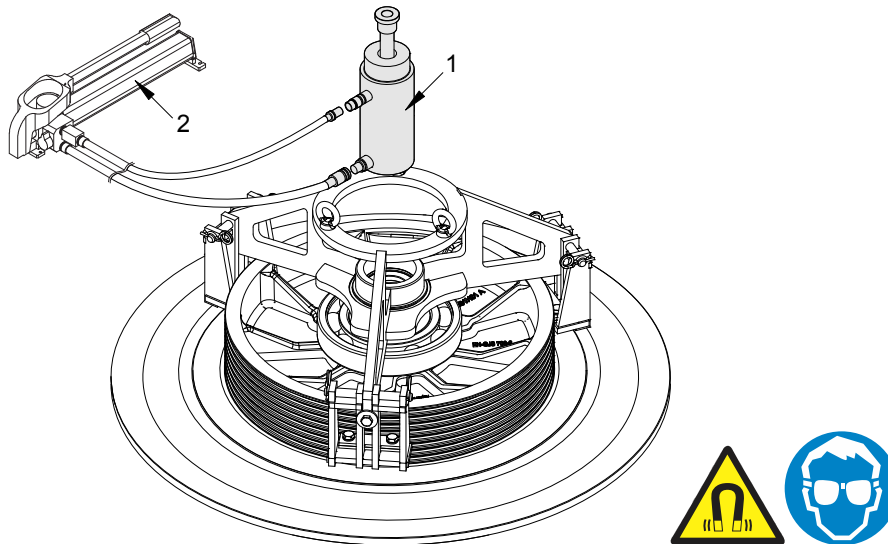
Always wear eye protection when using a hydraulic jack.

WARNING:

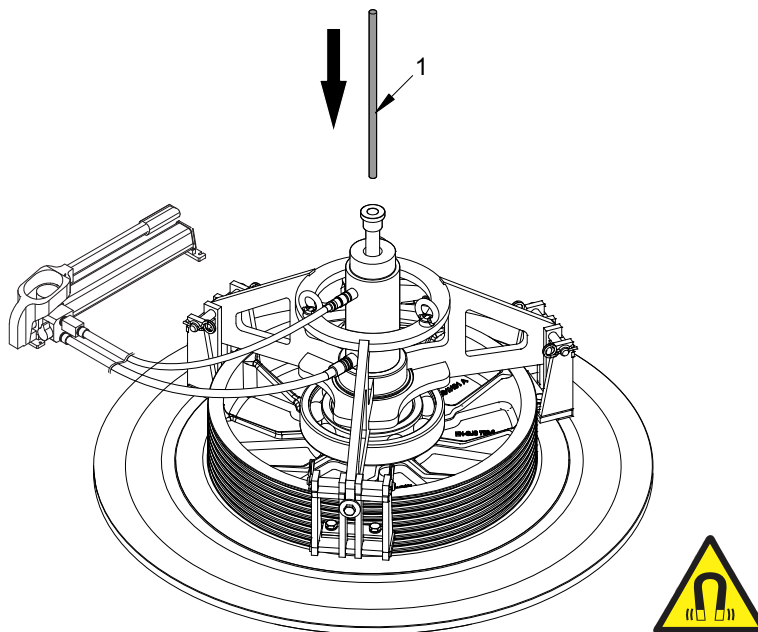
The rotor has a strong magnetic field and a strong attraction force. Handle the rotor carefully and take care not to get your hands crushed between the magnet and the tools.

NOTE:

Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.



5. Check tool alignment by inserting the M20 threaded bar (1).
 - Adjust if necessary.



6. Pump the piston out of the hydraulic jack (2) and lower the rotor assembly (1) in place with the brake in the hand pump.
- Keep load chains (3) on while lowering with the hand pump.
 - Gradually lower the rotor assembly (1) with both the chains and the pump.

WARNING:

Always wear eye protection when using a hydraulic jack.

NOTE:

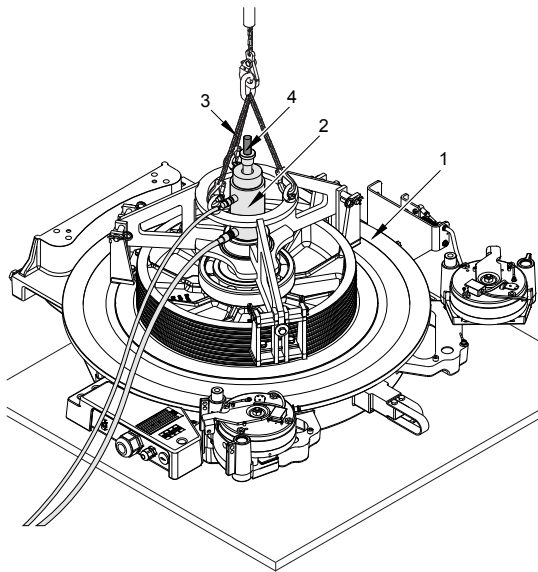
For more information on lower the rotor assembly correctly, refer to related information.

NOTE:

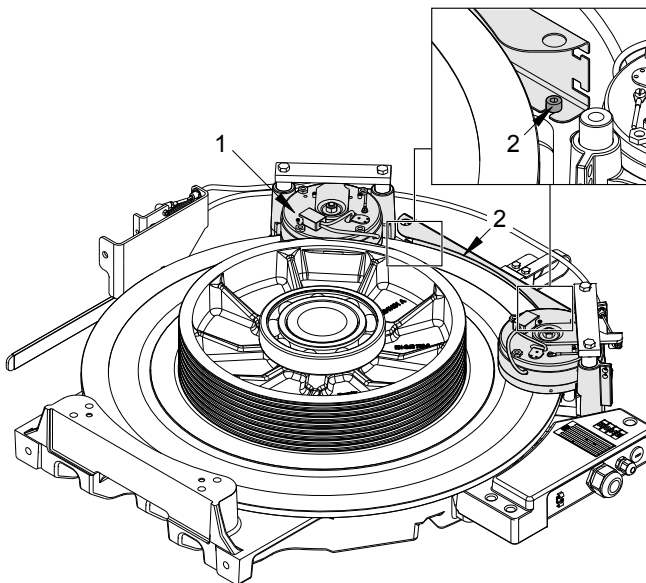
Make sure that hydraulic hoses run straight from hydraulic jack to hand pump without sharp bends.

NOTE:

Pull the rotor assembly (1) in place with threaded bar (4) and hydraulic jack (2) if needed.



7. Install the brakes (1) in place, slide the top cover (2) back and tighten the screws.
For more information, refer to related information.

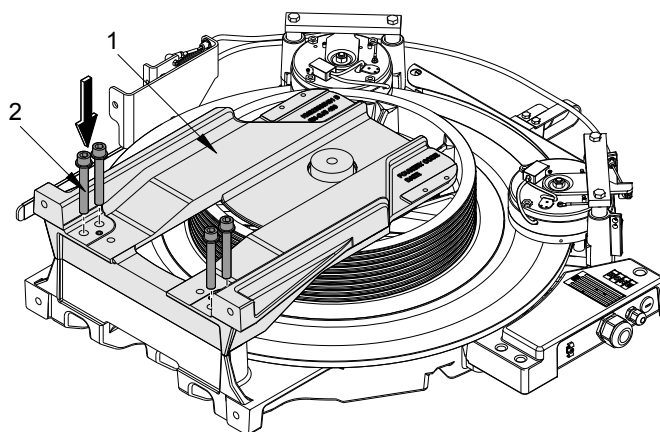


57665a71

8. Install the body2 (1) with bolts (2).
– Check that body2 (1) attaches evenly at the bottom and to the bearing fit area.

CAUTION:

Do not knock the body 2 in place with a hammer. Check if there is any blocking object if it does not fit completely in place.

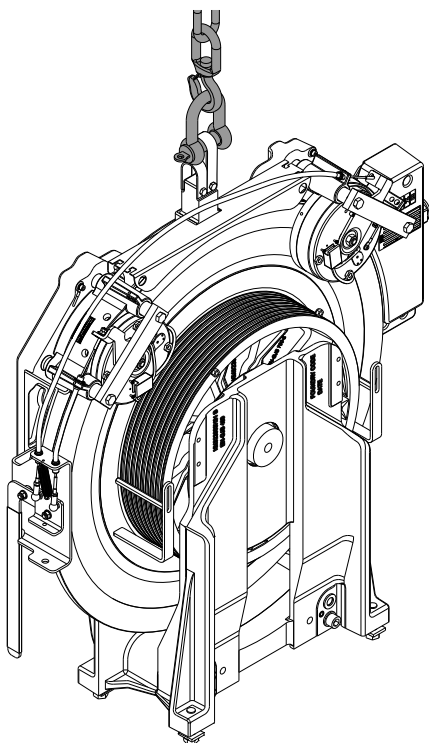


9960fdd0

9. Lift the machine until it stands up.

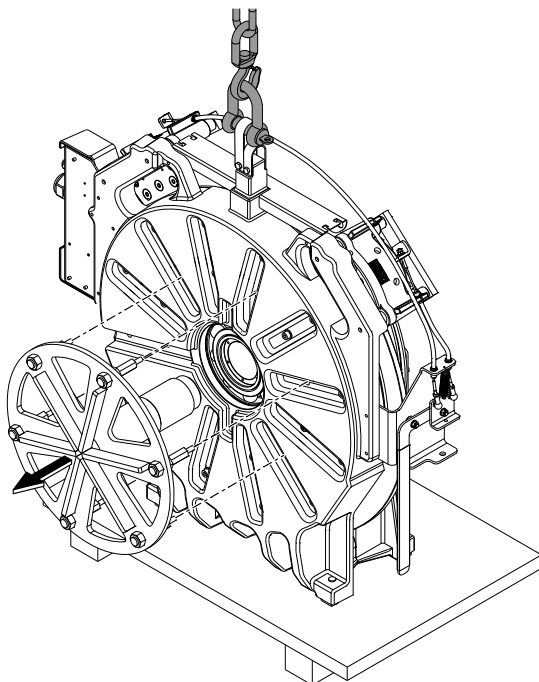
NOTE:

Always keep the chain hoist attached when the machinery is in vertical position.



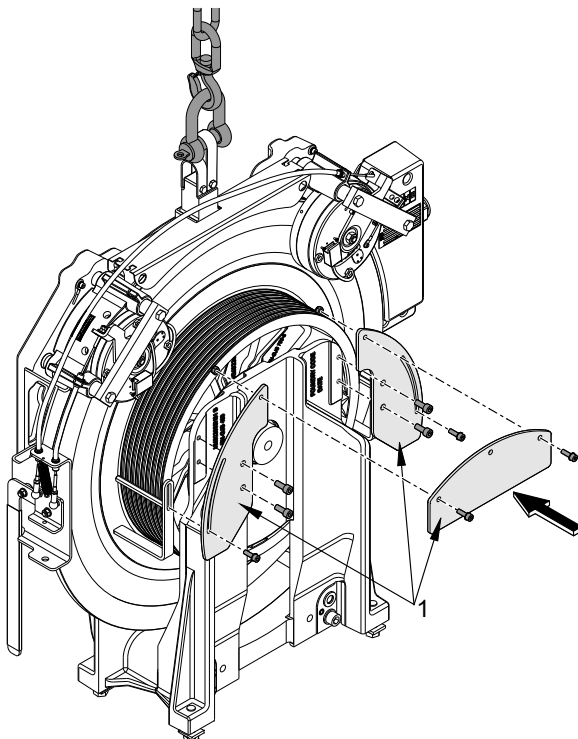
b52274e9

10. Remove the lower extractor tool.



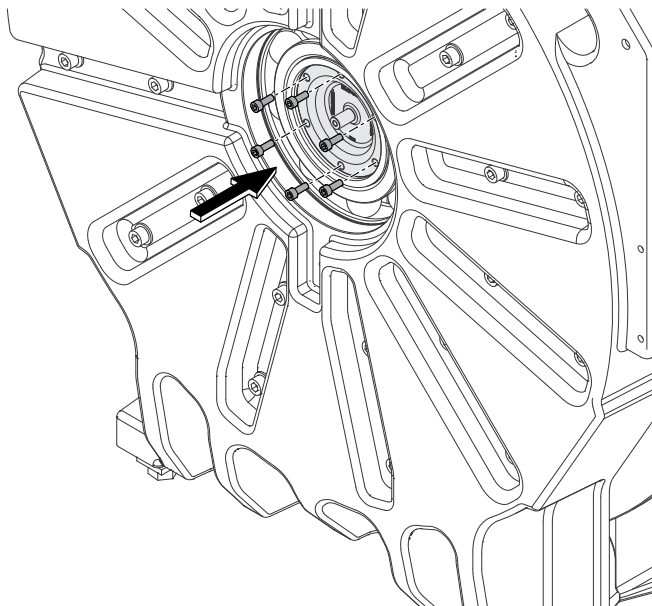
056463e9

11. Install the rope guards (1).



e6791005

12. Install the encoder elevator shaft flange.



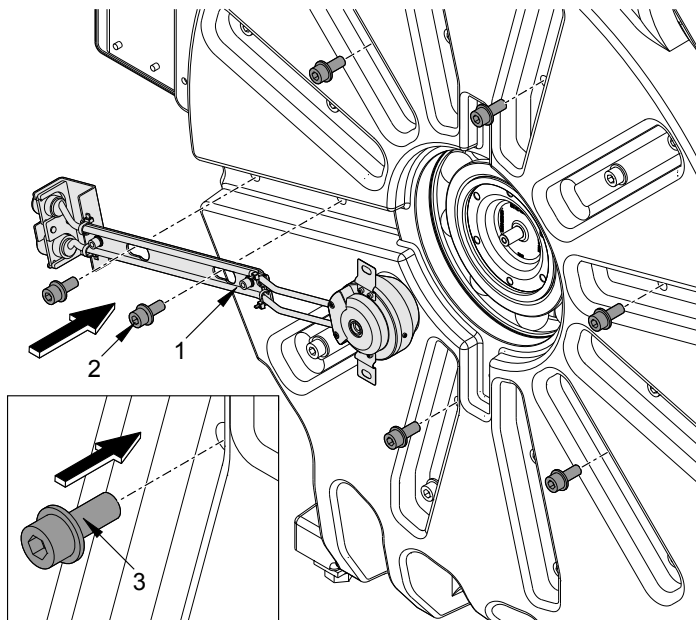
65c71c92

13. Install encoder wire holder plate (1) and the other hex socket screws (2, 3).

- Replace the M10 × 25 screws if needed.
- Tighten the M10 × 25 screws to 49 Nm.

CAUTION:

Use only hex socket screws M10 × 25. Longer screws damage the machine.

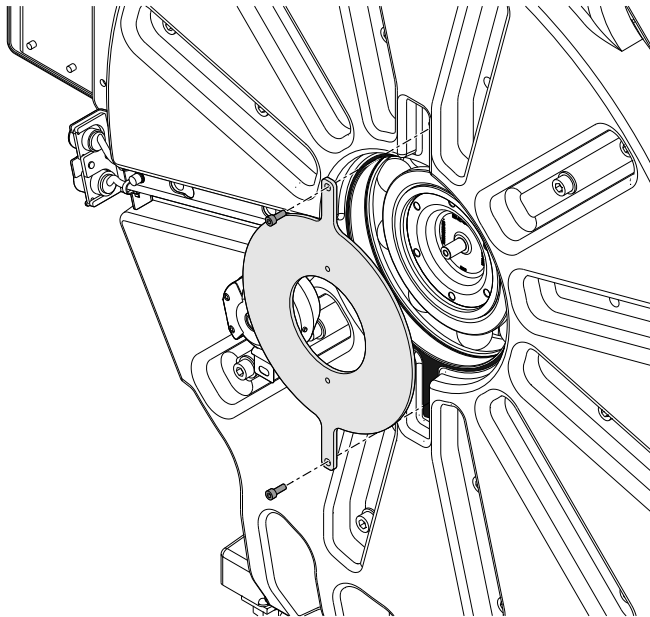


be1cff3d

14. Apply new silicone glue between the cast surface and the cover plate. Install the cover plate.

NOTE:

Do not exceed the amount of glue used before. Check that all old glue is removed at first.

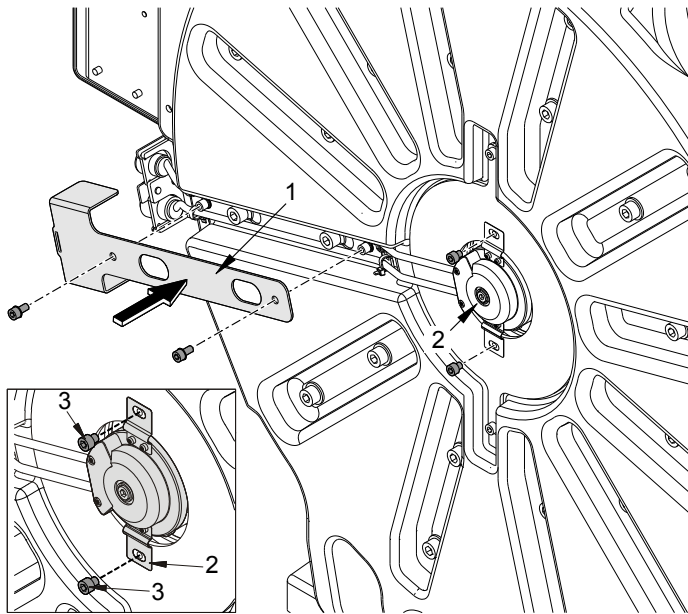


4e55a13b

15. Install washers between the encoder wire cover plate (1) and the cast surface. Install the encoder (2) and the encoder wire cover plate (1).

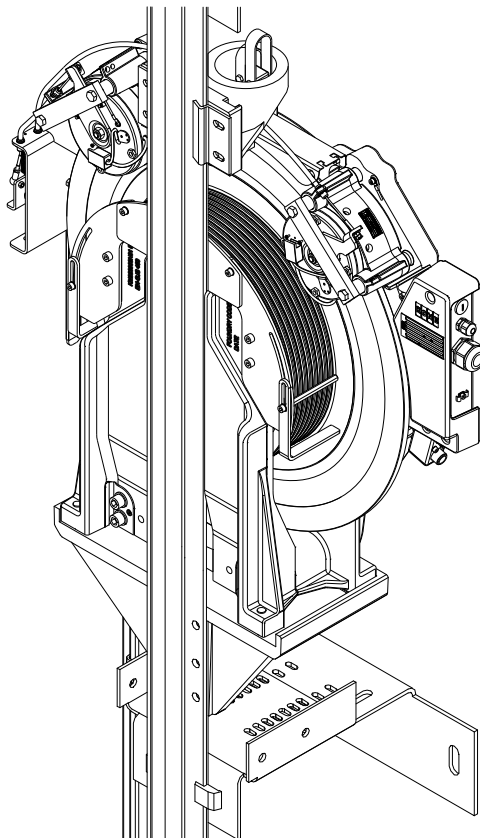
NOTE:

If encoder fixing screws (3) are replaced, a new thread locking compound must be added. Use Loctite 243 or equivalent.



16. Install the machine back to the elevator shaft / machine room.

- For more information on elevator with machine room, refer to the related information.
- For more information on machine-room-less elevator, refer to the related information.
- Test the brakes, for more information, refer to the related information.



Related information

- [*AS-04.06.001 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room-less Elevator*](#)
- [*AS-04.06.002 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room Elevator*](#)
- [*Remove brake unit \(96\)*](#)
- [*Use hydraulic jack with hand pump \(105\)*](#)

70126861

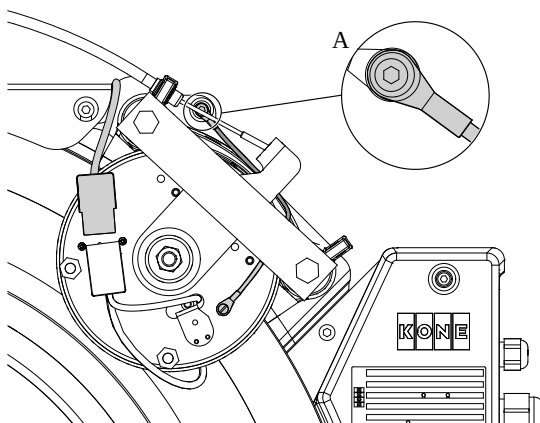
APPENDIX A. REMOVE BRAKE UNIT



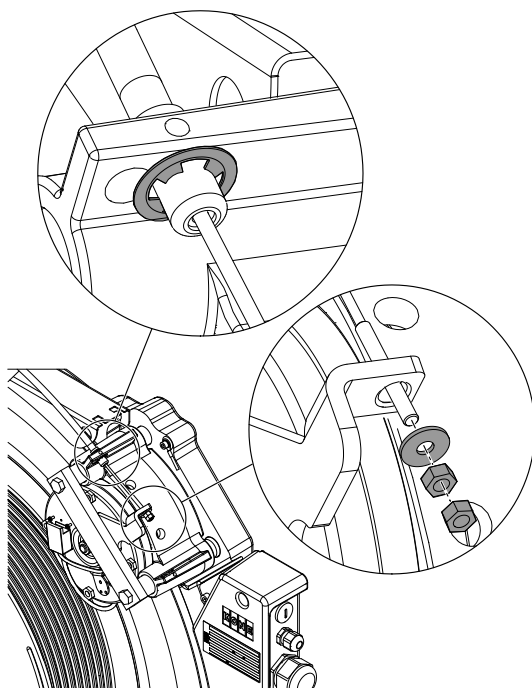
WARNING:

Do not remove both brakes simultaneously.

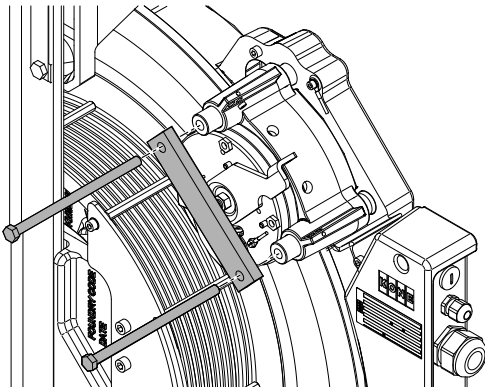
1. Disconnect the brake cable.
2. Disconnect the brake earthing wire (A).



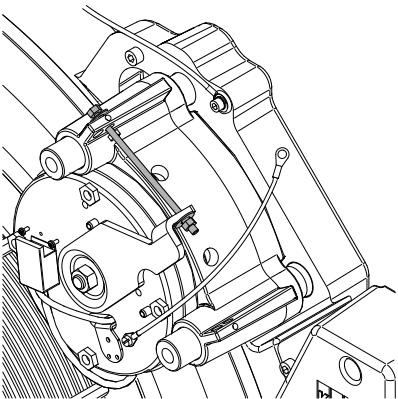
3. Disconnect the brake release cable from the brake.



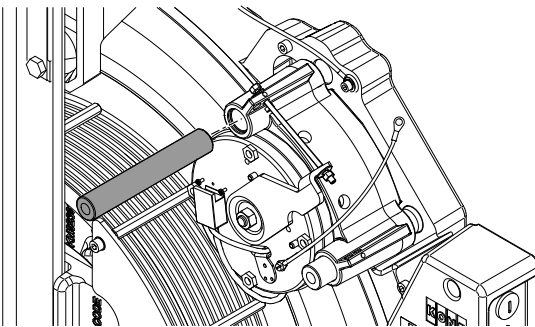
4. Remove the brake fixing bolts and the support plate.



5. Open the brake with M6x125 mm bolt, nut and washers (2 pcs.).



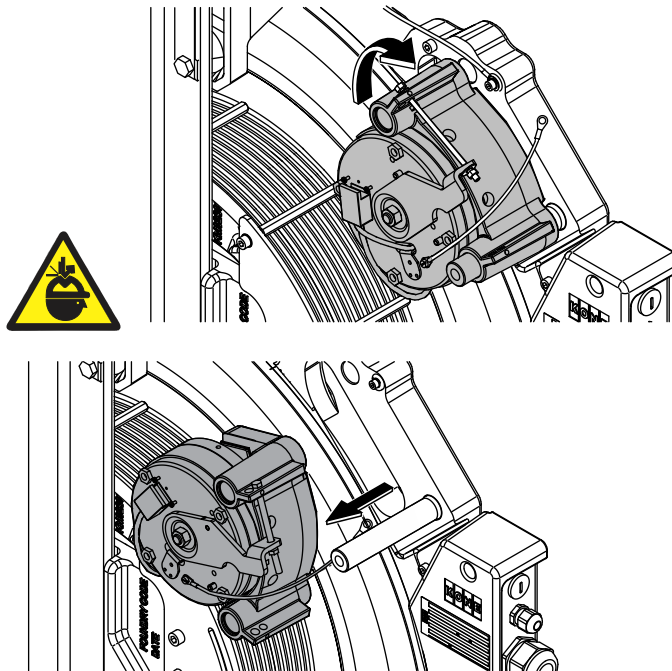
6. Remove the upper pin while holding the brake firmly.



NOTE:

The upper end of the brake becomes loose when the pin is removed.

7. Turn the brake upwards and pull it out.



NOTE:

If the brake unit does not move freely, clean the brake pin. Check for possible damage and use fine grade sand paper or emery cloth to remove the scratches.

Make sure that metal particles cannot get into the machine.

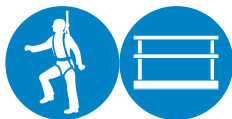
-
8. Place the brake on a solid surface.

NOTE:

Make sure that nothing gets inside the brake when the brake closes.

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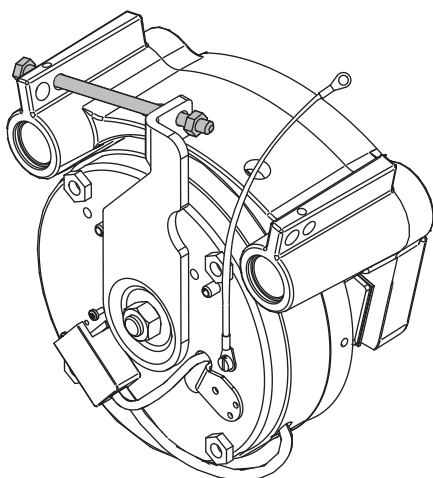
APPENDIX B. INSTALL AND ADJUST BRAKE UNIT



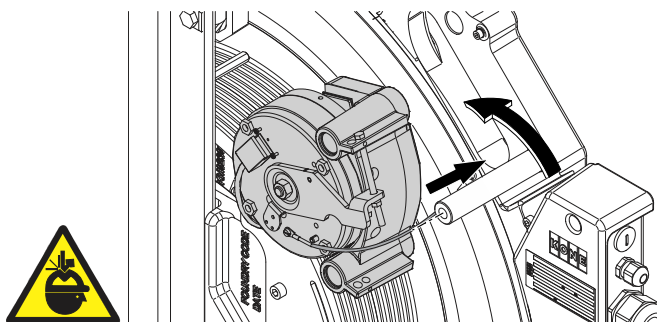
NOTE:

The maintenance person must verify CE marking in the safety component, store the Declaration of Conformity document in the logbook folder in the site, and enter the safety component changes in the elevator logbook.

1. Open the new brake with M6x125 mm bolt, nut and washers (2 pcs.).



2. Install the brake to the lower brake pin and slide it inwards.



NOTE:

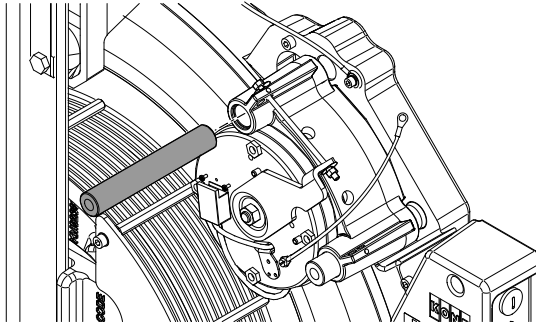
Check that the new brake unit is correct type and does not have transportation damage.

3. Turn the brake downwards until the brake disc is between the brake pads.
-

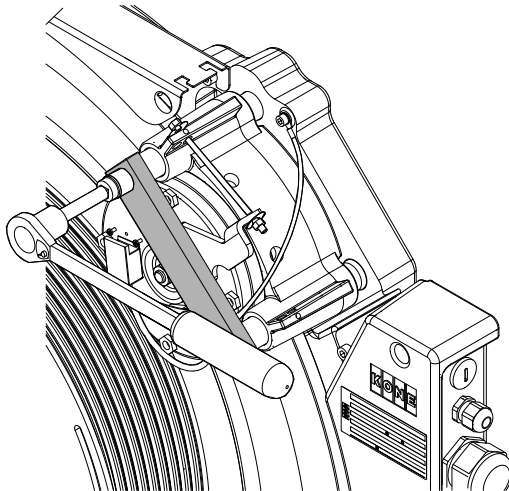
NOTE:

Do not the damage the brake pads against the brake disc.

4. Install the upper brake pin.



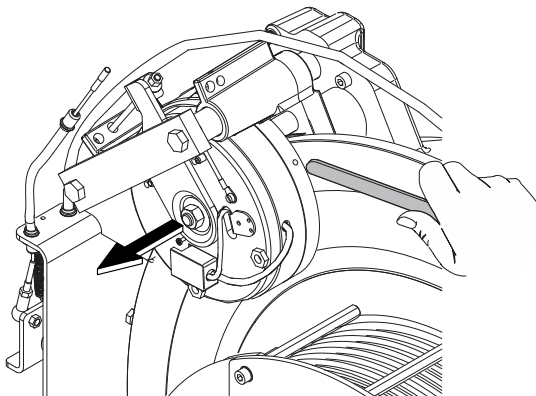
5. Install the brake support plate and the brake bolts.



NOTE:

Tighten the brake bolts to 72 Nm.

6. Move the brake body by hand.
- The brake must move slightly on the brake pins only when the brake is open (this is the clearance from the air gap).
- If the brake is open and it does not move freely, check the brake fixing.
7. Move the brake by hand until the entire air gap is on one side of the brake disc and the other brake pad touches the brake disc.

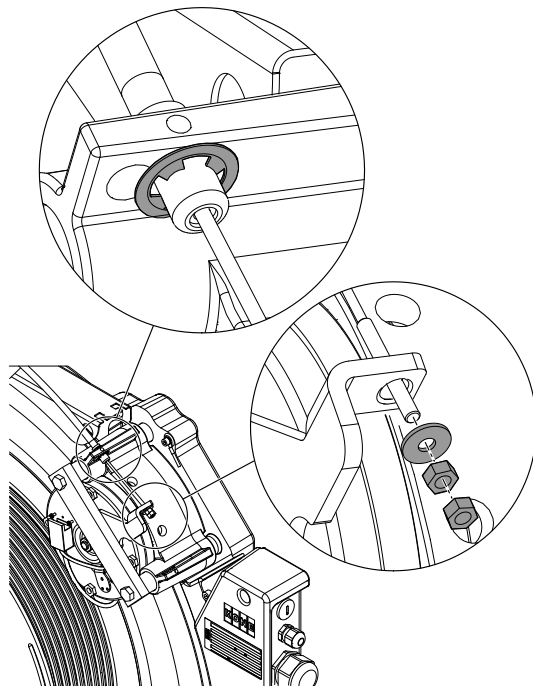


Measure the air gap with 0.2 mm and 0.6 mm feeler gauges (for example, 0.5 + 0.1 mm). The 0.2 mm feeler gauge must fit, 0.6 mm must not fit (exact limit is 0.55 mm).

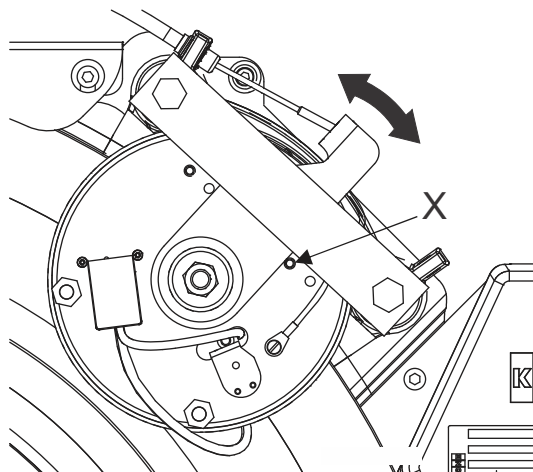
NOTE:

If the brake does not pass this test, the brake must be replaced.

-
8. Remove the bolt, nut and washers (2 pcs.). Connect and adjust the brake release cable.



9. Check that the brake lever rests on the limiter pin (X).



Move the brake lever by hand. The lever must move easily for at least 5 mm.

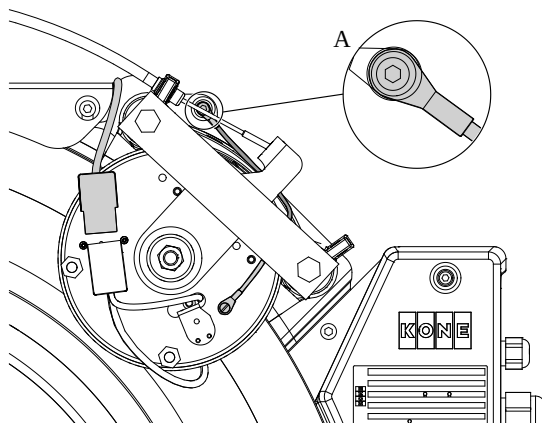
NOTE:

This check makes sure that brake spring force goes to the friction pads, not too manual opening mechanism.

10. Connect the brake earthing wire (A).

NOTE:

Check earth continuity. See section Measure protective earth continuity.



11. Connect the brake cable.

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APPENDIX C. PERFORM ONE-SIDED ELECTRICAL BRAKE TEST WITH 0% LOAD (LCE)

1. Make a car call to the topmost floor from the Maintenance Access Panel (MAP) / controller.
2. Switch on the elevator to Recall Drive Feature (RDF) (270).
3. Set the value of brake test parameter to 21.
This value activates the brake 1 test for one start only.
Brake test parameter:
 - KDL16, KDL32 and KDM: 6_72
 - V3F16R: 6_80
 - V3F16L and V3F18: Not available
4. Drive the elevator on RDF upwards.
To drive RDF upwards, push RDF 'RUN' and 'UP' buttons simultaneously.

NOTE:

The motor does not try to run.

If the elevator car moves, stop the test (release the RDF RUN and UP buttons).

Brake 2 opens, brake 1 remains closed for testing.

The drive stops the test by itself after 10 seconds (maximum).

5. Check the drive fault code from the error log.

NOTE:

Drive fault code 126, subcode 6022:

- Test passed.

Drive fault code 126 and subcode 2071 or 2072:

- Test failed.
-

If the elevator car moves or drive fault codes are displayed, brake 1 did not pass the test. Check the following possible causes for brake 1:

- Brake shoe is mechanically stuck.
- Incorrect counterweight / car balance
- Dirt on the brake shoe
- External lubricant leakage
- Internal lubricant leakage
- Wrong brake cable connection in the motor connection box.

Repeat the test. If the test fails again, replace the brakes or adjust the brakes.

6. Set the value of brake test parameter to 22.
This value activates the brake 2 test for one start only.
7. Drive the elevator on RDF upwards.
To drive RDF upwards, hold the RDF RUN and UP buttons together.

NOTE:

The motor does not try to run.

If the elevator car moves, stop the test (release the RDF RUN and UP buttons).

Brake 1 opens, brake 2 remains closed for testing.

The drive stops the test by itself after 10 seconds (maximum).

8. Check the drive fault code from the error log.

NOTE:

Drive fault code 126, subcode 6021:

- Test passed.

Drive fault code 126 and subcode 2071 or 2072:

- Test failed.

If the elevator car moves or drive fault codes are displayed, brake 2 did not pass the test. Check the following possible causes for brake 2:

- Brake shoe is mechanically stuck.
- Incorrect counterweight / car balance
- Dirt on the brake shoe
- External lubricant leakage
- Internal lubricant leakage
- Wrong brake cable connection in the motor connection box.

Repeat the test. If the test fails again, replace the brakes or adjust the brakes.

9. Switch the elevator to normal drive (270).

Related information

- [*AS-04.06.001 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room-less Elevator*](#)
- [*AS-04.06.002 Machine and Brake Replacement Instruction for MX14 Machine in Machine-room Elevator*](#)

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APPENDIX D. USE HYDRAULIC JACK WITH HAND PUMP

WARNING:



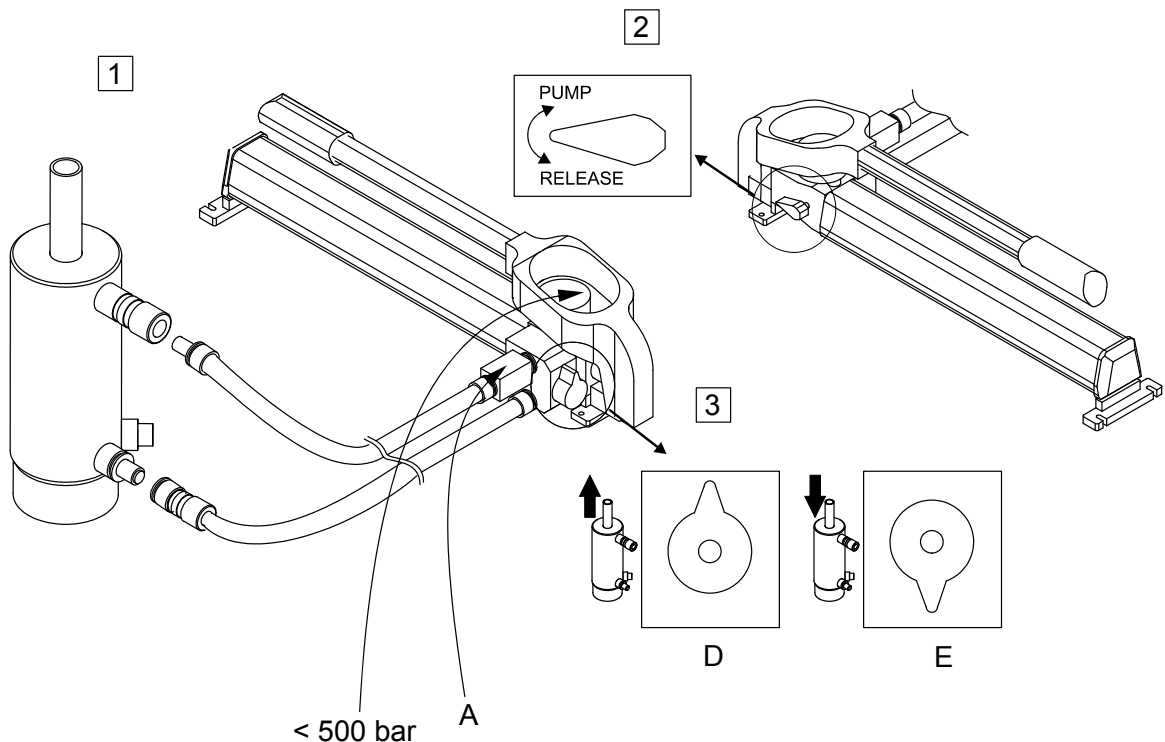
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Risk of high-pressure oil leakage when using the hydraulic pump. Use eye protection.

CAUTION:

Use the hand pump as far away from the hydraulic jack as the length of the connected hydraulic hoses allow. Keep the hydraulic jack in sight during operation.

1. Connect the hydraulic hoses.
2. Use the valve.
When the valve is in the up position, the pump is pressurizing.
When the valve is in the down position, pressure is released.
3. Select the moving direction of the piston.



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WARNING:

Do not exceed pressure of 500 bar. Make sure that the pressure is within the allowed working range specified on the type label.

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